Templates

Group 1 — Product Definition

- Product & Requirements (PRD)
- User Research & Discovery (URD)
- Stakeholders & Governance (STK)
- Domain Model & Glossary (DMG)
- Roadmap & Scope Control (RSC)
- Risk & Assumptions (RISK)
- Business Rules & Policy (BRP)
- Success Metrics & Instrumentation Plan (SMIP)

Group 2 — Experience Design

- Design & UX/UI (DES)
- Interaction Design & Motion (IXD)
- Content Design & UX Writing (CDX)
- Design System & UI Tokens (DSYS)
- Information Architecture & Navigation (IAN)
- Accessibility Design (A11YD)
- Responsive Layout & Breakpoints (RLB)
- Visual Asset Production (VAP)

Group 3 — System Architecture

- Architecture & Contracts (ARC)
- System Interfaces & Integration Contracts (SIC)
- Service Boundaries & Deployment Topology (SBDT)
- Permission Model & Authorization Design (PMAD)
- Error Model & Reason Codes (ERR)
- Realtime & Messaging Architecture (RTM)
- Workflow & Orchestration Design (WFO)
- API Governance & Versioning (APIG)

Group 4 — Data & Information

- Data Model & Schema (DATA)
- Data Lifecycle & Retention (DLR)
- Data Governance & Lineage (DGL)
- Data Quality & Validation (DQV)
- Search & Indexing (SRCH)
- Caching & Data Access Patterns (CACHE)

- Reporting & Aggregations (RPT)

Group 5 — Application Build

- Backend/API (API)
- Background Jobs & Scheduling (JBS)
- Eventing & Webhooks (EVT)
- Rate Limits & Abuse Controls (RLIM)
- Feature Flags & Config (FFCFG)
- API Pagination/Filtering/Sorting (PFS)
- File Processing & Media Pipelines (FPMP)
- Admin & Internal Tools APIs (ADMIN)
- Frontend/UI Implementation (FE)
- State Management & Data Fetching (SMD)
- Client Performance & Rendering (CPR)
- Forms & Validation (FORM)
- Client Routing & Deep Links (ROUTE)
- UI Composition & Layout Patterns (UICP)
- Client Error Handling & Recovery (CER)
- Client Security (CSec)
- Mobile Implementation (MOB)
- Mobile Device Capabilities (MDC)
- Offline & Sync (OFS)
- Mobile Performance & Battery (MBAT)
- Mobile Deep Links & Universal Links (MDL)
- Push Notifications (MPUSH)
- App Store Release & Signing (SIGN)

Group 6 — Integrations & External Services

- Integration & External Systems (INT)
- Third-Party Auth & SSO (SSO)
- CRM/ERP Integrations (CRMERP)
- Webhooks Consumers & Providers (WHCP)
- Payments & Billing (PAY)
- Notifications & Comms (NOTIF)
- Files/Media & Storage (FMS)

Group 7 — Security, Privacy, Compliance

- Security & Privacy (SEC)
- Identity & Access (IAM)

- Threat Modeling & Abuse Prevention (TMA)
- Secrets & Key Management (SKM)
- Privacy Modeling & Data Minimization (PRIV)
- Audit Logging & Forensics (AUDIT)
- Compliance & Risk (COMP)

Group 8 — Operations & Reliability

- Observability (OBS)
- Analytics & Telemetry (ANL)
- Logging & Tracing Standards (LTS)
- Monitoring & Alerting (ALRT)
- SLOs/SLAs & Error Budgets (SLO)
- Performance & Scalability (PERF)
- Load/Stress Planning (LOAD)
- Capacity & Cost Modeling (COST)
- Performance Budgets & Profiling (PBP)
- Quality & Testing (QA)
- QA Automation & Harnesses (QAH)
- Regression & Journey Tests (RJT)
- Test Data & Environments (TDE)
- Reliability & Resilience (RELIA)
- Incident Response & Postmortems (IRP)
- DevOps & Deployment (OPS)
- Infrastructure as Code (IAC)
- CI/CD Pipelines (CICD)
- Release & Change Management (REL)
- Environment Management (ENV)
- Secrets Deployment & Rotation (SDR)
- Backup/Restore & DR (BDR)
- Documentation & Runbooks (DOC)
- Localization & Accessibility (L10N/A11Y)

Product Definition

Product Definition + Physical Build Coverage — Template Set (Locked v2)

Product & Requirements (PRD)
PRD-01 Product Overview
PRD-02 Goals, Non-Goals, Success Metrics
PRD-03 Personas & Roles
PRD-04 Feature Catalog (by ID)
PRD-05 User Stories / JTBD Map
PRD-06 Non-Functional Requirements (NFRs)
PRD-07 Constraints (constraints only; assumptions live in RISK)
PRD-08 Open Questions Log
PRD-09 Acceptance Criteria Catalog (by feature/flow)

User Research & Discovery (URD)
URD-01 Research Plan (questions, methods, sample)
URD-02 Findings Summary (themes + evidence)
URD-03 User Needs & Pain Points (ranked)
URD-04 Journey Map (current vs target)
URD-05 Validation Plan (what to test, how to measure)

Stakeholders & Governance (STK)
STK-01 Stakeholder Map (roles, decision rights)
STK-02 Decision Log (what/why/when)
STK-03 RACI / Ownership Matrix
STK-04 Approval Gates (who signs what, when)

Domain Model & Glossary (DMG)
DMG-01 Domain Glossary (canonical terms)
DMG-02 Concept Model (entities + relationships, narrative)
DMG-03 Invariants & Definitions (must-always-be-true rules)
DMG-04 Event Vocabulary (canonical events/actions)

Roadmap & Scope Control (RSC)
RSC-01 Release Roadmap (milestones)
RSC-02 Scope Boundaries (in/out + rationale)
RSC-03 Prioritization Framework (method + scoring)
RSC-04 Change Control Policy (how scope changes)

Risk & Assumptions (RISK)
RISK-01 Assumptions Register
RISK-02 Risk Register (probability/impact/mitigation)
RISK-03 Dependency Map (external/internal)
RISK-04 Contingency Triggers (what causes plan changes)

Business Rules & Policy (BRP)
BRP-01 Business Rules Catalog (by ID)
BRP-02 Eligibility & Entitlement Rules
BRP-03 Pricing/Permission Policy Rules (if applicable)
BRP-04 Exceptions & Edge-Case Policy

Success Metrics & Instrumentation Plan (SMIP)
SMIP-01 KPI/OKR Definitions (targets + owners)
SMIP-02 Analytics Event Spec (event names + properties)
SMIP-03 Funnel/Conversion Definitions
SMIP-04 Experiment Measurement Plan (guardrails + success)

Product & Requirements

Product & Requirements (PRD)

- **PRD-01** Product Overview
- **PRD-02** Goals, Non-Goals, Success Metrics
- **PRD-03** Personas & Roles
- **PRD-04** Feature Catalog (by ID)
- **PRD-05** User Stories / JTBD Map
- **PRD-06** Non-Functional Requirements (NFRs)
- **PRD-07** Constraints & Assumptions
- **PRD-08** Open Questions Log
- **PRD-09** Acceptance Criteria Catalog (by feature/flow)

PRD-01

PRD-01 — Product Overview

Header Block

- template_id: PRD-01
- title: Product Overview
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-01_Product_Overview.md
- compliance_gate_id: TMP-05.PRIMARY.PROD
- upstream_dependencies: []
- inputs_required: ["SPEC_INDEX", "DOMAIN_MAP", "GLOSSARY", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create the single, canonical overview of the product: what it is, who it serves, the problems it solves, and the boundaries of the build. This document must be consistent with the Canonical Spec and must not invent requirements not present in upstream inputs.

Inputs Required

- SPEC_INDEX: {{spec.index}}
- DOMAIN_MAP: {{domain.map}} | OPTIONAL
- GLOSSARY: {{glossary.terms}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing docs/notes: {{inputs.notes}} | OPTIONAL

Required Fields

- Product name
- One-sentence product summary
- Primary user types
- Problem statement
- Value proposition
- Primary use cases (top 3–7)
- Scope boundaries (in-scope / out-of-scope)
- Success definition (high level)

- Constraints (top 3–10)
- Assumptions (top 3–10)

Optional Fields

- Competitive/alternative solutions | OPTIONAL
- Business model overview | OPTIONAL
- Phasing notes (MVP vs later) | OPTIONAL
- Key risks | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Do not restate full details that belong in other PRD docs; reference them.
- Every claim must be traceable to an input or explicitly marked **UNKNOWN**.
- Do not introduce new feature IDs; use: {{spec.features_by_id}} and reference IDs (feat_*) as given.
- If a required field is not provided by inputs, write **UNKNOWN** unless explicitly disallowed.
- Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1) Product Identity

- **Name:** {{product.name}}
- **Tagline / 1-liner:** {{product.oneliner}}
- **Summary:** {{product.summary_paragraph}}

2) Users & Audience

- **Primary user types:**
  - {{spec.personas_by_id[p1].name}} | OPTIONAL
  - {{spec.personas_by_id[p2].name}} | OPTIONAL
- **Secondary user types:** {{product.secondary_users}} | OPTIONAL

3) Problem Statement

- **Problem:** {{product.problem_statement}}
- **Why now:** {{product.why_now}} | OPTIONAL
- **Current workaround(s):** {{product.current_workarounds}} | OPTIONAL

4) Value Proposition

- **Primary value:** {{product.primary_value}}
- **Differentiators:** {{product.differentiators}} | OPTIONAL

5) Core Use Cases (Top)

List the top use cases in priority order. Each should map to one or more feature IDs.

1. {{use_cases[0].name}} — {{use_cases[0].summary}}
    - Features: {{use_cases[0].feature_ids}} (e.g., {{spec.features_by_id[feat_x].name}})
2. {{use_cases[1].name}} — {{use_cases[1].summary}}
    - Features: {{use_cases[1].feature_ids}}
3. {{use_cases[2].name}} — {{use_cases[2].summary}}
    - Features: {{use_cases[2].feature_ids}}

6) Scope Boundaries

- **In-scope (MVP):**
    - {{scope.in[0]}}
    - {{scope.in[1]}}
- **Out-of-scope (explicit):**
    - {{scope.out[0]}}
    - {{scope.out[1]}}
- **Future / Later (non-commit):** {{scope.later}} | OPTIONAL

7) Success Definition (High Level)

- **What “success” means:** {{success.definition}}
- **Primary success signals:**
    - {{success.signals[0]}}
    - {{success.signals[1]}}
- **Metrics pointer:** {{xref:SMIP-01}} | OPTIONAL

8) Constraints

List constraints that materially shape the build (technical, business, legal, time).

- {{constraints[0]}}
- {{constraints[1]}}
- {{constraints[2]}}

9) Assumptions

List assumptions that must hold for the plan to work.

- {{assumptions[0]}}
- {{assumptions[1]}}
- {{assumptions[2]}}

10) Risks (Optional)

- {{risks[0]}} | OPTIONAL
- {{risks[1]}} | OPTIONAL

11) Open Questions (Optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

12) References

- Feature catalog: {{xref:PRD-04}}
- NFRs: {{xref:PRD-06}}
- Business rules: {{xref:BRP-01}} | OPTIONAL
- Metrics plan: {{xref:SMIP-01}} | OPTIONAL
- Domain map: {{domain.map}} | OPTIONAL

Cross-References

- Upstream: {{xref:Canonical_Spec}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
- Downstream: {{xref:PRD-04}}, {{xref:PRD-06}}, {{xref:IMP-01}} | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Fill fields with **UNKNOWN** where needed; do not invent.
- intermediate: Required. Replace **UNKNOWN** with sourced statements when inputs exist.
- advanced: Required. Add crisp boundary statements and traceability pointers to SPEC IDs.

Unknown Handling

- UNKNOWN_ALLOWED: `why_now`, `current_workarounds`, `differentiators`, `secondary_users`, `scope.later`, `risks`, `open_questions`

- If any Required Field is UNKNOWN, allow only if: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
- If UNKNOWN appears in scope boundaries, must be flagged in “Open Questions”.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.PROD
- Pass conditions:
    - required_fields_present == true
    - all feature references use existing IDs (no new IDs introduced)
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

PRD-02

PRD-02 — Goals, Non-Goals, Success Metrics

Header Block

- template_id: PRD-02
- title: Goals, Non-Goals, Success Metrics
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-02_Goals_NonGoals_Success.md
- compliance_gate_id: TMP-05.PRIMARY.PROD
- upstream_dependencies: ["PRD-01"]
- inputs_required: ["SPEC_INDEX", "PRD-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Lock the product's intended outcomes and boundaries: what success looks like, what is explicitly not being attempted, and the measurable signals that indicate progress. This prevents scope creep and makes later planning, testing, and release gates objective.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- SPEC_INDEX: {{spec.index}}
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing KPI notes: {{inputs.kpi_notes}} | OPTIONAL
- Existing roadmap/milestones: {{inputs.roadmap}} | OPTIONAL

Required Fields

- Goals (3–10)
- Non-Goals (3–15)
- Success metrics (3–12)
- Metric definitions (name, description, formula/logic, target)
- Measurement cadence
- Ownership (who tracks each metric)
- Time horizon (MVP / 30-60-90 / release)

Optional Fields

- Guardrail metrics | OPTIONAL
- Baseline values (current) | OPTIONAL
- Leading vs lagging classification | OPTIONAL
- Segment definitions (per persona / tier) | OPTIONAL
- Data source notes | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Goals must be outcome-oriented; features do not qualify as goals.
- Non-goals must be explicit and testable (what will not be built / supported).
- Metrics must be measurable; if not measurable, mark **UNKNOWN** and create an instrumentation question.
- If a metric depends on analytics events, reference: {{xref:SMIP-02}} | OPTIONAL
- Do not invent baselines/targets; if missing, mark **UNKNOWN** (unless standards disallow).
- If a goal has no metric mapping, it fails the Completeness Gate.

Output Format

1) Goals

List outcomes the product must achieve.

goal_id	goal_statement	primary_user_type	mapped_feature_ids	notes
goal_01	{{goals[0].statement}}	{{goals[0].primary_user}}	{{goals[0].feature_ids}}	{{goals[0].notes}}
goal_02	{{goals[1].statement}}	{{goals[1].primary_user}}	{{goals[1].feature_ids}}	{{goals[1].notes}}
goal_03	{{goals[2].statement}}	{{goals[2].primary_user}}	{{goals[2].feature_ids}}	{{goals[2].notes}}

2) Non-Goals

List explicit exclusions to prevent scope creep.

non_goal_id	non_goal_statement	rationale	revisit_trigger
ng_01	{{non_goals[0].statement}}	{{non_goals[0].rationale}}	{{non_goals[0].revisit_trigger}}
ng_02	{{non_goals[1].statement}}	{{non_goals[1].rationale}}	{{non_goals[1].revisit_trigger}}
ng_03	{{non_goals[2].statement}}	{{non_goals[2].rationale}}	{{non_goals[2].revisit_trigger}}

3) Success Metrics (Core)

metric_id	metric_name	type (leading/lagging)	definition	formula/logic	target	time_horizon	cadence	owner
met_01	{{metrics[0].name}}	{{metrics[0].type}}	{{metrics[0].definition}}	{{metrics[0].formula}}	{{metrics[0].target}}	{{metrics[0].time_horizon}}	{{metrics[0].cadence}}	{{metrics[0].owner}}
met_02	{{metrics[1].name}}	{{metrics[1].type}}	{{metrics[1].definition}}	{{metrics[1].formula}}	{{metrics[1].target}}	{{metrics[1].time_horizon}}	{{metrics[1].cadence}}	{{metrics[1].owner}}
met_03	{{metrics[2].name}}	{{metrics[2].type}}	{{metrics[2].definition}}	{{metrics[2].formula}}	{{metrics[2].target}}	{{metrics[2].time_horizon}}	{{metrics[2].cadence}}	{{metrics[2].owner}}

4) Goal → Metric Mapping (Required)

Every goal must map to at least one metric.

goal_id	mapped_metric_ids	pass_condition
goal_01	{{goal_metric_map[goal_01].metric_ids}}	{{goal_metric_map[goal_01].pass_condition}}
goal_02	{{goal_metric_map[goal_02].metric_ids}}	{{goal_metric_map[goal_02].pass_condition}}

goal_03	{{goal_metric_map[goal_03].metric_ids}}	{{goal_metric_map[goal_03].pass_condition}}

5) Guardrails (Optional)

Metrics that must not degrade while optimizing core success.

metric_id	metric_name	definition	threshold	owner
gr_01	{{guardrails[0].name}}	{{guardrails[0].definition}}	{{guardrails[0].threshold}}	{{guardrails[0].owner}}

6) Measurement Notes

- **Primary data sources:** {{measurement.data_sources}} | OPTIONAL
- **Segmentation rules:** {{measurement.segments}} | OPTIONAL
- **Attribution/identity assumptions:** {{measurement.attribution}} | OPTIONAL

7) Open Questions (Optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
- Downstream: {{xref:SMIP-01}} | OPTIONAL, {{xref:SMIP-02}} | OPTIONAL, {{xref:PRD-04}}, {{xref:QA-01}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Targets/baselines may be **UNKNOWN**; mappings must still exist.
- intermediate: Required. Replace **UNKNOWN** with sourced targets where inputs exist; add cadence/owner for each metric.
- advanced: Required. Add pass/fail thresholds per metric and explicit guardrails; ensure metric logic is implementation-ready.

Unknown Handling

- UNKNOWN_ALLOWED: `baseline_values`, `targets`, `data_sources`, `segments`, `attribution`, `guardrails`, `open_questions`
- If a goal lacks any measurable metric → block Completeness Gate (no UNKNOWN exception).

Completeness Gate

- Gate ID: TMP-05.PRIMARY.PROD
- Pass conditions:
    - required_fields_present == true
    - every_goal_has_metric_mapping == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

PRD-03

PRD-03 — Personas & Roles

Header Block

- template_id: PRD-03
- title: Personas & Roles
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-03_Personas_Roles.md
- compliance_gate_id: TMP-05.PRIMARY.PROD
- upstream_dependencies: ["PRD-01"]
- inputs_required: ["SPEC_INDEX", "PRD-01", "GLOSSARY", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": false}

Purpose

Define the canonical user personas and system roles used throughout the kit. Personas describe user intent; roles describe permissions and access. This doc establishes consistent naming, privileges, and role-to-feature mapping for downstream design, API authorization, and testing.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- SPEC_INDEX: {{spec.index}}
- GLOSSARY: {{glossary.terms}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing org notes: {{inputs.org_notes}} | OPTIONAL

Required Fields

- Personas list (2–12)
- For each persona: name, description, goals, pain points, primary workflows
- Roles list (1–20)
- For each role: role name, description, permissions summary, scope
- Persona ↔ Role mapping (who typically has which role)
- Role ↔ Feature access mapping (high level)
- Role hierarchy / precedence (if applicable)

Optional Fields

- Persona segments (tiers, plans) | OPTIONAL
- Anti-personas / excluded users | OPTIONAL
- Role request/approval flow | OPTIONAL
- Role lifecycle (invited, active, suspended) | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Do not invent features; reference existing IDs from: {{spec.features_by_id}}
- Roles are authorization constructs; personas are research constructs. Do not merge them.
- Permissions must be described in plain language; implementation details belong in IAM/ARC docs.
- If a persona or role is uncertain, mark **UNKNOWN** and list in Open Questions.
- Use consistent terminology from: {{glossary.terms}} | OPTIONAL

Output Format

1) Personas (canonical)

persona_id	name	summary	primary_goals	main_pain_points	primary_workflows	notes
p1	{{personas[p1].name}}	{{personas[p1].summary}}	{{personas[p1].goals}}	{{personas[p1].pain_points}}	{{derive:PERSONA_TO_WORKFLOWS(p1)}}	{{personas[p1].notes}}
p2	{{personas[p2].name}}	{{personas[p2].summary}}	{{personas[p2].goals}}	{{personas[p2].pain_points}}	{{derive:PERSONA_TO_WORKFLOWS(p2)}}	{{personas[p2].notes}}

2) Roles (canonical)

role_id	role_name	description	scope	permissions_summary	typical_personas	notes

r_admin	{{roles[r_admin].name}}	{{roles[r_admin].description}}	{{roles[r_admin].scope}}	{{roles[r_admin].permissions}}	{{roles[r_admin].typical_personas}}	{{roles[r_admin].notes}}
r_user	{{roles[r_user].name}}	{{roles[r_user].description}}	{{roles[r_user].scope}}	{{roles[r_user].permissions}}	{{roles[r_user].typical_personas}}	{{roles[r_user].notes}}

3) Persona ↔ Role Mapping (required)

persona_id	typical_role_ids	rationale
p1	{{persona_role_map[p1].role_ids}}	{{persona_role_map[p1].rationale}}
p2	{{persona_role_map[p2].role_ids}}	{{persona_role_map[p2].rationale}}

4) Role Hierarchy / Precedence (required if applicable)

- **Higher overrides lower:** {{role_hierarchy.order}} | OPTIONAL
- **Conflicts resolved by:** {{role_hierarchy.conflict_rule}} | OPTIONAL

5) Role ↔ Feature Access (high level)

For each role, list allowed feature IDs and restricted feature IDs.

role_id	allowed_feature_ids	restricted_feature_ids	notes
r_admin	{{role_feature_access[r_admin].allowed}}	{{role_feature_access[r_admin].restricted}}	{{role_feature_access[r_admin].notes}}
r_user	{{role_feature_access[r_user].allowed}}	{{role_feature_access[r_user].restricted}}	{{role_feature_access[r_user].notes}}

6) Role Lifecycle (optional)

- **States:** {{role_lifecycle.states}} | OPTIONAL
- **How roles are granted:** {{role_lifecycle.grant_flow}} | OPTIONAL
- **How roles are revoked:** {{role_lifecycle.revoke_flow}} | OPTIONAL

7) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:SPEC_INDEX}} | OPTIONAL
- Downstream: {{xref:IAM-*}}* | *OPTIONAL, {{xref:ARC-04}}* | *OPTIONAL, {{xref:SEC-}}* | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Keep permissions_summary high-level; allow UNKNOWN where inputs are missing.
- intermediate: Required. Add role_feature_access for all roles; minimize UNKNOWN by deriving from feature list.
- advanced: Not required. (Advanced detail goes into IAM/ARC authorization specs.)

Unknown Handling

- UNKNOWN_ALLOWED: `role_hierarchy`, `role_lifecycle`, `anti_personas`, `segments`, `open_questions`
- If a role has no permissions_summary → block Completeness Gate (no UNKNOWN exception).

Completeness Gate

- Gate ID: TMP-05.PRIMARY.PROD
- Pass conditions:
  - required_fields_present == true
  - personas_count >= 2
  - roles_count >= 1
  - every_role_has_permissions_summary == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

PRD-04

PRD-04 — Feature Catalog (by ID) + Acceptance Hooks

Header Block

- template_id: PRD-04
- title: Feature Catalog (by ID) + Acceptance Hooks
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-04_Feature_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.REQ
- upstream_dependencies: ["PRD-01", "PRD-03"]
- inputs_required: ["SPEC_INDEX", "PRD-01", "PRD-03", "GLOSSARY", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the authoritative, machine-addressable catalog of product features using stable feature IDs. This is the anchor for planning, design, implementation, and test coverage. Each feature includes acceptance hooks that point downstream to flows, screens, endpoints, and test cases (without redefining those details here).

Inputs Required

- SPEC_INDEX: {{spec.index}}
- PRD-01: {{xref:PRD-01}}
- PRD-03: {{xref:PRD-03}} | OPTIONAL
- GLOSSARY: {{glossary.terms}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing backlog: {{inputs.backlog}} | OPTIONAL

Required Fields

- Feature list (minimum 5 for non-trivial products; allow smaller only if product is small)
- For each feature:
    - feature_id (stable)
    - name

    - description
    - primary persona(s)
    - priority (P0/P1/P2)
    - status (planned/in_progress/done)
    - dependencies (feature_ids)
    - success signals (high level)
    - acceptance hooks (pointers: flows/screens/endpoints/tests) with placeholders allowed
- Feature grouping (epics/modules) OR domain mapping (optional but recommended)

Optional Fields

- Feature exclusions (not in MVP) | OPTIONAL
- Feature risks | OPTIONAL
- Effort estimate (t-shirt sizing) | OPTIONAL
- Rollout strategy | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Feature IDs must be stable and unique. Do not renumber once assigned.
- Use this format for IDs unless constrained otherwise: `feat_<short_slug>` (e.g., `feat_auth_login`).
- Do not invent personas/roles; reference: {{spec.personas_by_id}} | OPTIONAL, {{xref:PRD-03}} | OPTIONAL
- Acceptance hooks are pointers only; details live in downstream docs (DES/ARC/API/QA).
- If a required feature attribute is unknown, mark **UNKNOWN** and add to Open Questions.
- Every P0 feature must have at least one acceptance hook in each of:
    - Flow OR Screen (DES)
    - Endpoint OR Component (ARC/API/FE)
    - Test case (QA)
      If not available yet, placeholders are allowed but must be present.

Output Format

1) Feature Index (summary)

feature_id	name	priority	primary_personas	status	depends_on

{{spec.features_by_id[feat_x].id}}	{{spec.features_by_id[feat_x].name}}	{{features[feat_x].priority}}	{{features[feat_x].personas}}	{{features[feat_x].status}}	{{features[feat_x].depends_on}}
{{spec.features_by_id[feat_y].id}}	{{spec.features_by_id[feat_y].name}}	{{features[feat_y].priority}}	{{features[feat_y].personas}}	{{features[feat_y].status}}	{{features[feat_y].depends_on}}

2) Feature Details (required, one block per feature)

{{spec.features_by_id[feat_x].id}} — {{spec.features_by_id[feat_x].name}}

- **Description:** {{features[feat_x].description}}
- **Primary personas:** {{features[feat_x].personas}}
- **Priority:** {{features[feat_x].priority}} (P0/P1/P2)
- **Status:** {{features[feat_x].status}} (planned/in_progress/done)
- **Dependencies:** {{features[feat_x].depends_on}} | OPTIONAL
- **Success signals:** {{features[feat_x].success_signals}} | OPTIONAL

Acceptance Hooks (pointers)

- Flows: {{features[feat_x].acceptance_hooks.flow_ids}} (e.g., {{xref:DES-01}} → flow_*) | OPTIONAL
- Screens: {{features[feat_x].acceptance_hooks.screen_ids}} (e.g., {{xref:DES-04}} → screen_*) | OPTIONAL
- Endpoints: {{features[feat_x].acceptance_hooks.endpoint_ids}} (e.g., {{xref:ARC-02}} → ep_*) | OPTIONAL
- Components: {{features[feat_x].acceptance_hooks.component_ids}} (e.g., {{xref:DES-03}} → comp_*) | OPTIONAL
- Test cases: {{features[feat_x].acceptance_hooks.test_case_ids}} (e.g., {{xref:QA-02}} → tc_*) | OPTIONAL

Derived mappings (optional)

- Workflows: {{derive:FEATURE_TO_WORKFLOWS(feat_x)}} | OPTIONAL
- Domains impacted: {{features[feat_x].domains}} | OPTIONAL

Notes / Risks (optional)

- {{features[feat_x].notes}} | OPTIONAL
- {{features[feat_x].risks}} | OPTIONAL

---

3) Feature Grouping (optional)

group_id	group_name	feature_ids
grp_01	{{feature_groups[0].name}}	{{feature_groups[0].feature_ids}}

4) MVP Exclusions (optional)

excluded_feature_id	rationale	revisit_trigger
{{excluded[0].feature_id}}	{{excluded[0].rationale}}	{{excluded[0].revisit_trigger}}

5) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-03}} | OPTIONAL, {{xref:SPEC_INDEX}} | OPTIONAL
- Downstream: {{xref:DES-01}}, {{xref:DES-04}}, {{xref:ARC-02}}, {{xref:IMP-01}}, {{xref:QA-01}}, {{xref:QA-02}}
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Provide feature list + P0/P1/P2 + basic acceptance hook placeholders.
- intermediate: Required. Add dependencies and domain impacts; reduce UNKNOWN; ensure P0 hooks include tests.
- advanced: Required. Add success signals and rollout notes; ensure hooks fully map into flows/screens/endpoints/tests.

Unknown Handling

- UNKNOWN_ALLOWED: `depends_on`, `success_signals`, `domains`, `notes`, `risks`, `feature_groups`, `excluded`, `open_questions`
- P0 features: acceptance hook placeholders are allowed, but absence is not.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.REQ
- Pass conditions:
    - required_fields_present == true
    - feature_ids_unique == true
    - p0_features_have_hooks == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

PRD-05

PRD-05 — User Stories / JTBD Map

Header Block

- template_id: PRD-05
- title: User Stories / JTBD Map
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-05_User_Stories_JTBD.md
- compliance_gate_id: TMP-05.PRIMARY.REQ
- upstream_dependencies: ["PRD-01", "PRD-03", "PRD-04"]
- inputs_required: ["SPEC_INDEX", "PRD-01", "PRD-03", "PRD-04", "GLOSSARY", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": false}

Purpose

Translate the feature catalog into user-centered intent statements (User Stories and/or Jobs-To-Be-Done) that can be traced to feature IDs and later to flows, screens, endpoints, and test cases. This creates a clear “why” layer without redefining implementation.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-03: {{xref:PRD-03}}
- PRD-04: {{xref:PRD-04}}
- SPEC_INDEX: {{spec.index}}
- GLOSSARY: {{glossary.terms}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Story/JTBD format choice (story, jtbd, or both)
- For each item:
    - story_id / job_id
    - persona_id
    - linked_feature_ids
    - intent statement (story or JTBD)
    - acceptance outcomes (plain language)

    - priority (P0/P1/P2)
    - edge cases summary
- Coverage mapping: every P0 feature has at least 1 story/job

Optional Fields

- Preconditions | OPTIONAL
- Frequency / importance | OPTIONAL
- Related workflows (derived) | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Do not create new features; map only to existing IDs from {{spec.features_by_id}}.
- Use canonical persona IDs from {{spec.personas_by_id}} or {{xref:PRD-03}}.
- If acceptance outcomes are unknown, mark **UNKNOWN** and add to Open Questions.
- P0 coverage is mandatory; if missing, fails Completeness Gate.

Output Format

1) Format Selection

- **Chosen formats:** {{stories.format}} (story / jtbd / both)
- **Story template:** “As a {{persona}}, I want {{capability}} so that {{benefit}}.”
- **JTBD template:** “When {{situation}}, I want to {{motivation}} so I can {{expected_outcome}}.”

2) User Stories (if used)

story_id	persona_id	story	linked_feature_ids	priority	acceptance_outcomes	edge_cases
us_01	{{stories[0].persona_id}}	{{stories[0].text}}	{{stories[0].feature_ids}}	{{stories[0].priority}}	{{stories[0].outcomes}}	{{stories[0].edge_cases}}
us_02	{{stories[1].persona_id}}	{{stories[1].text}}	{{stories[1].feature_ids}}	{{stories[1].priority}}	{{stories[1].outcomes}}	{{stories[1].edge_cases}}

3) JTBD Items (if used)

job_id	persona_id	situation	motivation	expected_outcome	linked_feature_ids	priority	edge_cases
job_01	{{jobs[0].persona_id}}	{{jobs[0].situation}}	{{jobs[0].motivation}}	{{jobs[0].expected_outcome}}	{{jobs[0].feature_ids}}	{{jobs[0].priority}}	{{jobs[0].edge_cases}}
job_02	{{jobs[1].persona_id}}	{{jobs[1].situation}}	{{jobs[1].motivation}}	{{jobs[1].expected_outcome}}	{{jobs[1].feature_ids}}	{{jobs[1].priority}}	{{jobs[1].edge_cases}}

4) P0 Feature Coverage (required)

feature_id	feature_name	has_story_or_job	mapped_story_ids / job_ids
{{spec.features_by_id[feat_p0].id}}	{{spec.features_by_id[feat_p0].name}}	{{coverage[feat_p0].covered}}	{{coverage[feat_p0].item_ids}}

5) Derived Workflow Mapping (optional)

- {{derive:FEATURE_TO_WORKFLOWS(feat_x)}} | OPTIONAL

6) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-03}}, {{xref:PRD-04}}
- Downstream: {{xref:DES-01}}, {{xref:DES-04}}, {{xref:QA-02}} | OPTIONAL, {{xref:IMP-01}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Provide at least one story/job per P0 feature; edge cases may be brief.
- intermediate: Required. Add acceptance outcomes that are testable; reduce UNKNOWN.
- advanced: Not required. (Advanced detail moves into flows/screens/test cases.)

Unknown Handling

- UNKNOWN_ALLOWED: `preconditions`, `frequency`, `derived_workflows`, `open_questions`, `edge_cases` (brief)
- If a P0 feature has no story/job mapping → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.REQ
- Pass conditions:
  - required_fields_present == true
  - every_p0_feature_covered == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

PRD-06

PRD-06 — Non-Functional Requirements (NFRs)

Header Block

- template_id: PRD-06
- title: Non-Functional Requirements (NFRs)
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-06_NFRs.md
- compliance_gate_id: TMP-05.PRIMARY.NFR
- upstream_dependencies: ["PRD-01", "PRD-02", "PRD-04"]
- inputs_required: ["SPEC_INDEX", "PRD-01", "PRD-02", "PRD-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the measurable non-functional requirements that constrain architecture, implementation, ops, and testing: performance, reliability, security, privacy, scalability, availability, maintainability, accessibility, and compliance. These must be written as testable requirements with thresholds wherever possible.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-02: {{xref:PRD-02}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- SPEC_INDEX: {{spec.index}}
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing SLOs/ops notes: {{inputs.slo_notes}} | OPTIONAL

Required Fields

- NFR categories included (at least: performance, reliability, security, privacy)
- For each NFR:
    - nfr_id
    - category

    - requirement statement
    - measurable threshold (or **UNKNOWN** if not available)
    - scope (system-wide / feature-specific / endpoint-specific)
    - verification method (how to test/measure)
    - owner
    - priority (P0/P1/P2)
- Explicit “MVP vs Later” notes for NFRs where applicable

Optional Fields

- Baseline (current) | OPTIONAL
- Dependencies (tools/infrastructure) | OPTIONAL
- Cost constraints | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- NFRs must be testable. If a threshold is unknown, include a verification plan for how it will be determined.
- Do not duplicate security controls; reference SEC templates for control catalogs.
- Every P0 NFR must have a verification method.
- If accessibility applies, include category “accessibility” and cross-reference L10N/A11Y templates.

Output Format

1) NFR Summary (by category)

category	count	p0_count	notes
performance	{{nfr_summary.performance.count}}	{{nfr_summary.performance.p0}}	{{nfr_summary.performance.notes}}
reliability	{{nfr_summary.reliability.count}}	{{nfr_summary.reliability.p0}}	{{nfr_summary.reliability.notes}}
security	{{nfr_summary.security.count}}	{{nfr_summary.security.p0}}	{{nfr_summary.security.notes}}
privacy	{{nfr_summary.privacy.count}}	{{nfr_summary.privacy.p0}}	{{nfr_summary.privacy.notes}}

2) NFR Catalog (required)

nfr_id	category	requirement	threshold	scope	verification_method	owner	priority	mvp_or_later
nfr_perf_01	performance	{{nfrs[0].requirement}}	{{nfrs[0].threshold}}	{{nfrs[0].scope}}	{{nfrs[0].verification}}	{{nfrs[0].owner}}	{{nfrs[0].priority}}	{{nfrs[0].phase}}
nfr_rel_01	reliability	{{nfrs[1].requirement}}	{{nfrs[1].threshold}}	{{nfrs[1].scope}}	{{nfrs[1].verification}}	{{nfrs[1].owner}}	{{nfrs[1].priority}}	{{nfrs[1].phase}}

3) Feature/Surface Mapping (optional but preferred)

Map NFRs to impacted features/endpoints/screens when not system-wide.

nfr_id	impacted_feature_ids	impacted_endpoint_ids	impacted_screen_ids	notes
nfr_perf_01	{{nfr_map[nfr_perf_01].feature_ids}}	{{nfr_map[nfr_perf_01].endpoint_ids}}	{{nfr_map[nfr_perf_01].screen_ids}}	{{nfr_map[nfr_perf_01].notes}}

4) Verification Plan Notes

- **Performance measurement plan:** {{verification.performance_plan}} | OPTIONAL
- **Reliability measurement plan:** {{verification.reliability_plan}} | OPTIONAL
- **Security validation plan:** {{verification.security_plan}} | OPTIONAL
- **Privacy validation plan:** {{verification.privacy_plan}} | OPTIONAL

5) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-02}} | OPTIONAL, {{xref:PRD-04}} | OPTIONAL
- Downstream: {{xref:ARC-01}}, {{xref:ARC-02}}, {{xref:SEC-01}}, {{xref:SEC-02}}, {{xref:QA-01}}, {{xref:PERF-*}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Thresholds may be **UNKNOWN** but must include verification_method.
- intermediate: Required. Add realistic thresholds where inputs exist; map NFRs to surfaces.
- advanced: Required. Add explicit SLO/SLA targets and error budget notes where applicable.

Unknown Handling

- UNKNOWN_ALLOWED: `threshold`, `baseline`, `dependencies`, `cost_constraints`, `open_questions`
- If priority == P0 and verification_method is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.NFR
- Pass conditions:
  - required_fields_present == true
  - nfr_categories_include_minimum == true
  - every_p0_nfr_has_verification_method == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

PRD-07

PRD-07 — Constraints & Assumptions

Header Block

- template_id: PRD-07
- title: Constraints & Assumptions
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-07_Constraints_Assumptions.md
- compliance_gate_id: TMP-05.PRIMARY.PROD
- upstream_dependencies: ["PRD-01", "PRD-06"]
- inputs_required: ["PRD-01", "PRD-06", "SPEC_INDEX", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Centralize the hard constraints and working assumptions that shape the build. This document prevents hidden requirements, forces explicit trade-offs, and feeds planning, architecture, risk management, and release gating.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- SPEC_INDEX: {{spec.index}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing constraints notes: {{inputs.constraints_notes}} | OPTIONAL

Required Fields

- Constraints list (minimum 5 unless truly small product)
- For each constraint:
    - constraint_id
    - statement
    - type (business/technical/legal/time/budget/platform)
    - rationale
    - impacted areas (domains/features)
    - enforcement (how it will be ensured)
    - severity (hard/soft)

- Assumptions list (minimum 5)
- For each assumption:
    - assumption_id
    - statement
    - rationale
    - risk if false
    - validation plan
    - owner
    - timeframe (when validated)

Optional Fields

- Linked NFRs | OPTIONAL
- Dependencies (external systems) | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Constraints are “must” statements; assumptions are “we believe” statements.
- Each assumption must include a validation plan (even if minimal).
- If enforcement or validation plan is unknown, mark **UNKNOWN** and add to Open Questions.
- If a constraint conflicts with another constraint/NFR, flag explicitly.

Output Format

1) Constraints Catalog (required)

constraint_id	statement	type	severity	rationale	impacted_domains	impacted_feature_ids	enforcement	notes
c_01	{{constraints[0].statement}}	{{constraints[0].type}}	{{constraints[0].severity}}	{{constraints[0].rationale}}	{{constraints[0].domains}}	{{constraints[0].feature_ids}}	{{constraints[0].enforcement}}	{{constraints[0].notes}}

c_02	{{constraints[1].statement}}	{{constraints[1].type}}	{{constraints[1].severity}}	{{constraints[1].rationale}}	{{constraints[1].domains}}	{{constraints[1].feature_ids}}	{{constraints[1].enforcement}}	{{constraints[1].notes}}

2) Assumptions Register (required)

assumption_id	statement	rationale	risk_if_false	validation_plan	owner	validate_by	notes
a_01	{{assumptions[0].statement}}	{{assumptions[0].rationale}}	{{assumptions[0].risk_if_false}}	{{assumptions[0].validation_plan}}	{{assumptions[0].owner}}	{{assumptions[0].validate_by}}	{{assumptions[0].notes}}
a_02	{{assumptions[1].statement}}	{{assumptions[1].rationale}}	{{assumptions[1].risk_if_false}}	{{assumptions[1].validation_plan}}	{{assumptions[1].owner}}	{{assumptions[1].validate_by}}	{{assumptions[1].notes}}

3) Conflicts / Tensions (required if any)

List explicit collisions (constraint vs constraint, constraint vs NFR, assumption vs constraint).

- {{conflicts[0]}} | OPTIONAL
- {{conflicts[1]}} | OPTIONAL

4) Derived Impact Summary (optional)

- Domains most impacted: {{derive:TOP_IMPACTED_DOMAINS(constraints, assumptions)}} | OPTIONAL
- Features most impacted: {{derive:TOP_IMPACTED_FEATURES(constraints, assumptions)}} | OPTIONAL

5) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-06}} | OPTIONAL
- Downstream: {{xref:RISK-01}} | OPTIONAL, {{xref:ARC-01}}, {{xref:IMP-01}}, {{xref:REL-*}} | OPTIONAL

- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Keep enforcement/validation simple; allow UNKNOWN only with open question.
- intermediate: Required. Add impacted_domains and validate_by dates; reduce UNKNOWN.
- advanced: Required. Add explicit conflict statements and mitigation notes tied to planning/release.

Unknown Handling

- UNKNOWN_ALLOWED: `impacted_feature_ids`, `enforcement` (soft constraints only), `notes`, `conflicts`, `open_questions`
- If an assumption.validation_plan is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.PROD
- Pass conditions:
  - required_fields_present == true
  - constraints_count >= 5 (unless explicitly justified)
  - assumptions_count >= 5 (unless explicitly justified)
  - every_assumption_has_validation_plan == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

PRD-08

PRD-08 — Open Questions Log

Header Block

- template_id: PRD-08
- title: Open Questions Log
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-08_Open_Questions_Log.md
- compliance_gate_id: TMP-05.PRIMARY.PROD
- upstream_dependencies: ["PRD-01", "PRD-04", "PRD-06", "PRD-07"]
- inputs_required: ["PRD-01", "PRD-04", "PRD-06", "PRD-07", "SPEC_INDEX", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Maintain a single canonical backlog of unresolved product questions that block or affect scope, design, architecture, security, testing, or release readiness. This prevents unknowns from being scattered across documents and enables deterministic gating (“no critical unknowns”).

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- PRD-07: {{xref:PRD-07}} | OPTIONAL
- SPEC_INDEX: {{spec.index}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Question list (can be 0, but must be explicit)
- For each question:
    - question_id
    - question
    - category (scope/design/architecture/data/security/ops/legal/metrics)
    - impact (P0 blocker / P1 important / P2 nice-to-have)
    - affected feature_ids / domains / docs

- owner
- status (open/answered/deferred)
- target_resolution_date (or **UNKNOWN**)
- resolution (if answered) OR next_step (if open)

Optional Fields

- Evidence links / notes | OPTIONAL
- Decision log pointer | OPTIONAL
- Dependencies (who/what is needed) | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL
- If any P0 blocker question is status=open, release gate must fail (downstream).
- Question IDs must be stable and unique (oq_01, oq_02… or oq_).
- If a question is “answered”, provide the resolution text and update affected docs list.
- Keep resolutions short; detailed answers belong in the relevant canonical doc and should be referenced.

Output Format

1) Open Questions Table (canonical)

question_id	question	category	impact	affected_feature_ids	affected_domains	affected_docs	owner	status	target_resolution_date	next_step_or_resolution
oq_01	{{questions[0].text}}	{{questions[0].category}}	{{questions[0].impact}}	{{questions[0].feature_ids}}	{{questions[0].domains}}	{{questions[0].docs}}	{{questions[0].owner}}	{{questions[0].status}}	{{questions[0].target_date}}	{{questions[0].next_or_resolution}}
oq_	{{question	{{questions[1]	{{questions[	{{questions[1].f	{{questions[1	{{question	{{questions	{{questions	{{questions[1].t	{{questions[1].next_

0 2	s[1].t ext}}	.categ ory}}	1].imp act}}	eature_ ids}}	].doma ins}}	s[1].d ocs}}	[1].ow ner}}	[1].st atus}}	arget_d ate}}	or_resolut ion}}

2) P0 Blockers Summary (required)

List all open P0 questions (if any).

- {{derive:LIST_P0_OPEN_QUESTIONS(questions)}} | OPTIONAL

3) Recently Answered (optional)

question_id	answered_on	resolution_summary	updated_docs
{{answered[0].id}}	{{answered[0].date}}	{{answered[0].resolution}}	{{answered[0].updated_docs}}

4) Notes (optional)

- {{notes[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-04}}, {{xref:PRD-06}}, {{xref:PRD-07}} | OPTIONAL
- Downstream: {{xref:STK-02}} | OPTIONAL, {{xref:RISK-01}} | OPTIONAL, {{xref:REL-*}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Owners and impacts may be coarse; keep list centralized.
- intermediate: Required. Add affected docs/domains; assign target dates where possible.
- advanced: Required. Maintain “answered” section with doc updates for traceability.

Unknown Handling

- UNKNOWN_ALLOWED: `target_resolution_date`, `dependencies`, `evidence_links`, `notes`
- If question.status == answered and resolution is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.PROD
- Pass conditions:
  - required_fields_present == true
  - question_ids_unique == true
  - answered_questions_have_resolution == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

PRD-09

PRD-09 — Acceptance Criteria Catalog (by Feature/Flow)

Header Block

- template_id: PRD-09
- title: Acceptance Criteria Catalog (by Feature/Flow)
- type: product_requirements
- template_version: 1.0.0
- output_path: 10_app/requirements/PRD-09_Acceptance_Criteria_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.REQ
- upstream_dependencies: ["PRD-04", "PRD-05", "PRD-06"]
- inputs_required: ["PRD-04", "PRD-05", "PRD-06", "SPEC_INDEX", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical acceptance criteria set that proves features are “done.” This catalog anchors QA planning and test case generation while remaining product-level (not implementation-level). Criteria are written in testable language and mapped to feature IDs and, when available, flow IDs.

Inputs Required

- PRD-04: {{xref:PRD-04}}
- PRD-05: {{xref:PRD-05}} | OPTIONAL
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- SPEC_INDEX: {{spec.index}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Acceptance criteria list (one or more per P0 feature)
- For each criterion:
    - ac_id
    - linked_feature_id
    - linked_flow_id (or **UNKNOWN**)

    - criterion statement (testable)
    - type (functional / validation / error-handling / security / performance / accessibility)
    - priority (P0/P1/P2)
    - pass condition (what must be true)
    - negative cases (at least one for P0 features)
- Coverage table: every P0 feature has >= 1 acceptance criterion

Optional Fields

- References to NFRs (nfr_id) | OPTIONAL
- References to business rules (br_id) | OPTIONAL
- Notes | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Criteria must be testable and unambiguous. Avoid “fast”, “easy”, “intuitive” unless quantified.
- For P0 features, include:
    - at least one happy-path criterion
    - at least one negative/edge criterion
- If a criterion depends on an NFR threshold, reference PRD-06 or nfr_id.
- Do not define UI layout; that belongs in DES templates. Reference screen IDs when needed.

Output Format

1) Acceptance Criteria Catalog (canonical)

ac_id	feature_id	flow_id	type	priority	criterion	pass_condition	negative_cases	refs (nfr/br/screen/endpoint)
ac_001	{{acceptance[0].feature_id}}	{{acceptance[0].flow_id}}	{{acceptance[0].type}}	{{acceptance[0].priority}}	{{acceptance[0].criterion}}	{{acceptance[0].pass_condition}}	{{acceptance[0].negative_cases}}	{{acceptance[0].refs}}

ac_002	{{acceptance[1].feature_id}}	{{acceptance[1].flow_id}}	{{acceptance[1].type}}	{{acceptance[1].priority}}	{{acceptance[1].criterion}}	{{acceptance[1].pass_condition}}	{{acceptance[1].negative_cases}}	{{acceptance[1].refs}}

2) P0 Feature Coverage (required)

feature_id	feature_name	acceptance_count	has_negative_case	mapped_ac_ids
{{spec.features_by_id[feat_p0].id}}	{{spec.features_by_id[feat_p0].name}}	{{coverage[feat_p0].count}}	{{coverage[feat_p0].has_negative}}	{{coverage[feat_p0].ac_ids}}

3) NFR-linked Criteria (optional)

nfr_id	related_ac_ids	notes
{{nfr_links[0].nfr_id}}	{{nfr_links[0].ac_ids}}	{{nfr_links[0].notes}}

4) Notes (optional)

- {{notes[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-04}}, {{xref:PRD-05}} | OPTIONAL, {{xref:PRD-06}} | OPTIONAL
- Downstream: {{xref:QA-01}}, {{xref:QA-02}}, {{xref:DES-01}} | OPTIONAL, {{xref:DES-04}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Ensure at least one criterion per P0 feature; flow_id may be UNKNOWN.
- intermediate: Required. Add pass_condition and negative cases for all P0 criteria.
- advanced: Required. Link criteria to NFRs and add clear refs for screens/endpoints where known.

Unknown Handling

- UNKNOWN_ALLOWED: `flow_id`, `refs`, `nfr_links`, `notes`
- If a P0 feature has zero acceptance criteria → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.REQ
- Pass conditions:
  - required_fields_present == true
  - every_p0_feature_has_acceptance == true
  - p0_acceptance_has_negative_cases == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

User Research & Discovery (URD)

User Research & Discovery (URD)

- **URD-01** Research Plan (questions, methods, sample)
- **URD-02** Findings Summary (themes + evidence)
- **URD-03** User Needs & Pain Points (ranked)
- **URD-04** Journey Map (current vs target)
- **URD-05** Validation Plan (what to test, how to measure)

URD-01

URD-01 — Research Plan (questions, methods, sample)

Header Block

- template_id: URD-01
- title: Research Plan (questions, methods, sample)
- type: user_research
- template_version: 1.0.0
- output_path: 10_app/research/URD-01_Research_Plan.md
- compliance_gate_id: TMP-05.PRIMARY.RESEARCH
- upstream_dependencies: ["PRD-01", "PRD-02", "PRD-03"]
- inputs_required: ["PRD-01", "PRD-02", "PRD-03", "GLOSSARY", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define a lightweight, execution-ready research plan that validates core product assumptions, clarifies user needs, and de-risks the highest-impact decisions. This plan is a guide for discovery work; it must remain aligned with product goals and avoid redefining requirements.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-02: {{xref:PRD-02}} | OPTIONAL
- PRD-03: {{xref:PRD-03}} | OPTIONAL
- GLOSSARY: {{glossary.terms}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing research notes: {{inputs.research_notes}} | OPTIONAL

Required Fields

- Research objectives (2–8)
- Key research questions (5–20)
- Hypotheses / assumptions under test (3–15)
- Methods (interviews, surveys, usability tests, diary study, etc.)
- Participant criteria (who qualifies)

- Sample size target (per method)
- Recruitment plan (how participants will be found)
- Script/guide outline (topics, not full script)
- Study logistics (tools, timeline, session length)
- Data capture plan (notes, recordings, consent)
- Analysis plan (how findings become themes)
- Output artifacts (what deliverables will be produced)

Optional Fields

- Incentives | OPTIONAL
- Risks / limitations | OPTIONAL
- Accessibility accommodations | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Research questions must map back to:
    - product goals: {{xref:PRD-02}}
    - personas/roles: {{xref:PRD-03}} | OPTIONAL
    - major assumptions: {{xref:RISK-01}} | OPTIONAL
- Do not claim results; this is a plan only.
- If sample size or recruitment approach is uncertain, mark **UNKNOWN** and include mitigation.
- Consent/recording must be explicitly addressed if any data is captured.

Output Format

1) Study Overview

- **Study name:** {{research.study_name}}
- **Owner:** {{research.owner}}
- **Stakeholders:** {{research.stakeholders}} | OPTIONAL
- **Timeline:** {{research.timeline}}
- **Session length:** {{research.session_length}}
- **Tools:** {{research.tools}} | OPTIONAL

2) Objectives

- {{research.objectives[0]}}
- {{research.objectives[1]}}

3) Key Questions

rq_id	question	mapped_goal_ids	mapped_persona_ids	mapped_assumption_ids
rq_01	{{research.questions[0].text}}	{{research.questions[0].goal_ids}}	{{research.questions[0].persona_ids}}	{{research.questions[0].assumption_ids}}
rq_02	{{research.questions[1].text}}	{{research.questions[1].goal_ids}}	{{research.questions[1].persona_ids}}	{{research.questions[1].assumption_ids}}

4) Hypotheses / Assumptions Under Test

h_id	hypothesis	why_it_matters	validation_signal
h_01	{{research.hypotheses[0].text}}	{{research.hypotheses[0].impact}}	{{research.hypotheses[0].signal}}
h_02	{{research.hypotheses[1].text}}	{{research.hypotheses[1].impact}}	{{research.hypotheses[1].signal}}

5) Methods (what + why)

method_id	method	purpose	sample_size_target	format	notes
m_01	{{research.methods[0].name}}	{{research.methods[0].purpose}}	{{research.methods[0].sample_size}}	{{research.methods[0].format}}	{{research.methods[0].notes}}

6) Participants

- **Target personas:** {{research.participants.target_persona_ids}} | OPTIONAL
- **Inclusion criteria:**
  - {{research.participants.inclusion[0]}}
  - {{research.participants.inclusion[1]}}
- **Exclusion criteria:**
  - {{research.participants.exclusion[0]}} | OPTIONAL
- **Recruitment channels:** {{research.participants.recruitment_channels}}
- **Incentives:** {{research.participants.incentives}} | OPTIONAL

7) Script / Guide Outline

- **Intro + consent**
- **Background questions**
- **Core tasks / prompts**
- **Concept reactions**
- **Wrap-up + follow-ups**
  (Outline bullets)
- {{research.guide_outline[0]}}
- {{research.guide_outline[1]}}

8) Logistics

- **Scheduling approach:** {{research.logistics.scheduling}}
- **Recording:** {{research.logistics.recording_policy}}
- **Consent handling:** {{research.logistics.consent_policy}}
- **Storage location:** {{research.logistics.storage_location}} | OPTIONAL
- **Accessibility accommodations:** {{research.logistics.accommodations}} | OPTIONAL

9) Analysis Plan

- **Approach:** {{research.analysis.approach}} (e.g., thematic analysis)
- **Coding scheme:** {{research.analysis.coding_scheme}} | OPTIONAL
- **How themes will be produced:** {{research.analysis.theme_process}}
- **How conflicts will be resolved:** {{research.analysis.conflict_resolution}} | OPTIONAL

10) Outputs / Deliverables

- URD-02 Findings Summary: {{xref:URD-02}}
- URD-03 Needs & Pain Points: {{xref:URD-03}}
- URD-04 Journey Map: {{xref:URD-04}} | OPTIONAL
- URD-05 Validation Plan: {{xref:URD-05}} | OPTIONAL

11) Risks / Limitations (optional)

- {{research.risks[0]}} | OPTIONAL
- {{research.risks[1]}} | OPTIONAL

12) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-02}} | OPTIONAL, {{xref:PRD-03}} | OPTIONAL, {{xref:RISK-01}} | OPTIONAL
- Downstream: {{xref:URD-02}}, {{xref:URD-03}}, {{xref:URD-04}} | OPTIONAL, {{xref:URD-05}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL, {{standards.rules[STD-PRIVACY]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Use 1–2 methods; keep plan short but complete.
- intermediate: Required. Map questions to goals/personas/assumptions; define analysis approach.
- advanced: Required. Add clear hypotheses and validation signals; include risk mitigation and rigor notes.

Unknown Handling

- UNKNOWN_ALLOWED: `sample_size_target`, `recruitment_channels`, `incentives`, `tools`, `storage_location`, `coding_scheme`, `accommodations`, `risks`, `open_questions`
- If data capture is planned and consent_policy is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESEARCH
- Pass conditions:
  - required_fields_present == true
  - questions_count >= 5
  - methods_count >= 1
  - participant_criteria_present == true
  - consent_policy_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

URD-02

URD-02 — Findings Summary (themes + evidence)

Header Block

- template_id: URD-02
- title: Findings Summary (themes + evidence)
- type: user_research
- template_version: 1.0.0
- output_path: 10_app/research/URD-02_Findings_Summary.md
- compliance_gate_id: TMP-05.PRIMARY.RESEARCH
- upstream_dependencies: ["URD-01"]
- inputs_required: ["URD-01", "PRD-01", "PRD-02", "PRD-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Capture research outcomes as clear themes supported by evidence. This document turns raw notes into actionable insights and creates traceable inputs for needs/pain points, journey mapping, PRD refinements, and risk reduction.

Inputs Required

- URD-01: {{xref:URD-01}}
- PRD-01: {{xref:PRD-01}} | OPTIONAL
- PRD-02: {{xref:PRD-02}} | OPTIONAL

- PRD-03: {{xref:PRD-03}} | OPTIONAL
- Raw notes/transcripts: {{inputs.raw_notes}} | OPTIONAL
- Recordings: {{inputs.recordings}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Study metadata (dates, methods used, sample)
- Participants summary (count + key segments)
- Themes list (minimum 3)
- For each theme:
    - theme_id
    - statement (1–2 sentences)
    - supporting evidence (quotes/observations)
    - frequency/strength signal (qualitative)
    - impacted personas
    - mapped research question IDs
    - implications (what it means for product)
- Top insights (top 3–7)
- Recommendations (what to do next) (3–10)
- Limitations / confidence notes

Optional Fields

- Contradictions / outliers | OPTIONAL
- Artifact links (notes repository) | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-PRIVACY]}} | OPTIONAL
- Evidence must be anonymized unless explicitly allowed.
- Themes must map back to URD-01 research questions where possible.
- Do not convert recommendations into requirements here; PRD updates happen downstream and must be tracked as decisions if changed.
- If evidence is missing for a theme, the theme cannot be included (fails gate).

Output Format

1) Study Metadata

- **Study name:** {{study.name}}
- **Dates:** {{study.dates}}
- **Methods used:** {{study.methods_used}}
- **Owner:** {{study.owner}}
- **Stakeholders:** {{study.stakeholders}} | OPTIONAL

2) Participants Summary

- **Total participants:** {{participants.total}}
- **Segments represented:** {{participants.segments}} | OPTIONAL

- **Primary personas:** {{participants.persona_ids}} | OPTIONAL
- **Recruitment notes:** {{participants.recruitment_notes}} | OPTIONAL

3) Themes (required)

theme_id	theme_statement	strength_signal	impacted_persona_ids	mapped_rq_ids	implications
th_01	{{themes[0].statement}}	{{themes[0].strength}}	{{themes[0].persona_ids}}	{{themes[0].rq_ids}}	{{themes[0].implications}}
th_02	{{themes[1].statement}}	{{themes[1].strength}}	{{themes[1].persona_ids}}	{{themes[1].rq_ids}}	{{themes[1].implications}}
th_03	{{themes[2].statement}}	{{themes[2].strength}}	{{themes[2].persona_ids}}	{{themes[2].rq_ids}}	{{themes[2].implications}}

4) Evidence Blocks (required, per theme)

th_01 — {{themes[0].statement}}

- **Evidence (anonymized):**
    - {{themes[0].evidence[0]}}
    - {{themes[0].evidence[1]}} | OPTIONAL
- **Notes:** {{themes[0].notes}} | OPTIONAL

th_02 — {{themes[1].statement}}

- **Evidence (anonymized):**
    - {{themes[1].evidence[0]}}
    - {{themes[1].evidence[1]}} | OPTIONAL

5) Top Insights (required)

1. {{insights[0]}}

2. {{insights[1]}}
3. {{insights[2]}}

6) Recommendations / Next Actions (required)

rec_id	recommendation	rationale	impacted_area (PRD/DES/ARC/etc.)	priority
rec_01	{{recs[0].text}}	{{recs[0].rationale}}	{{recs[0].area}}	{{recs[0].priority}}
rec_02	{{recs[1].text}}	{{recs[1].rationale}}	{{recs[1].area}}	{{recs[1].priority}}

7) Contradictions / Outliers (optional)

- {{contradictions[0]}} | OPTIONAL
- {{contradictions[1]}} | OPTIONAL

8) Limitations / Confidence (required)

- **Limitations:** {{limitations.text}}
- **Confidence level:** {{limitations.confidence}} (low/medium/high)
- **Why:** {{limitations.rationale}}

9) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:URD-01}}, {{xref:PRD-01}} | OPTIONAL, {{xref:PRD-02}} | OPTIONAL, {{xref:PRD-03}} | OPTIONAL

- Downstream: {{xref:URD-03}}, {{xref:URD-04}} | OPTIONAL, {{xref:URD-05}} | OPTIONAL, {{xref:PRD-04}} | OPTIONAL, {{xref:RISK-02}} | OPTIONAL, {{xref:STK-02}} | OPTIONAL

- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Minimum 3 themes with 1 evidence item each; keep recommendations concrete.

- intermediate: Required. Map themes to research questions and personas; add confidence notes.

- advanced: Required. Add contradictions/outliers and clearer implications + traceability to downstream changes.

Unknown Handling

- UNKNOWN_ALLOWED: `segments`, `stakeholders`, `recruitment_notes`, `artifact_links`, `contradictions`, `open_questions`

- If any theme has zero evidence items → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESEARCH

- Pass conditions:

    - required_fields_present == true

    - themes_count >= 3

    - every_theme_has_evidence == true

    - recommendations_count >= 3

- limitations_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

URD-03

URD-03 — User Needs & Pain Points (ranked)

Header Block

- template_id: URD-03
- title: User Needs & Pain Points (ranked)
- type: user_research
- template_version: 1.0.0
- output_path: 10_app/research/URD-03_Needs_PainPoints.md
- compliance_gate_id: TMP-05.PRIMARY.RESEARCH
- upstream_dependencies: ["URD-02"]
- inputs_required: ["URD-02", "PRD-03", "PRD-04", "GLOSSARY", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Convert research themes into a ranked catalog of user needs and pain points. This provides a deterministic input to prioritization, feature shaping, UX flows, and acceptance criteria without redefining implementation.

Inputs Required

- URD-02: {{xref:URD-02}}
- PRD-03: {{xref:PRD-03}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- GLOSSARY: {{glossary.terms}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Raw evidence links: {{inputs.evidence_links}} | OPTIONAL

Required Fields

- Needs list (minimum 5)
- Pain points list (minimum 5)
- For each item:
    - item_id
    - type (need / pain_point)

    - statement
    - impacted persona(s)
    - severity/importance (1–5 or low/med/high)
    - frequency signal (qualitative)
    - supporting evidence pointer(s)
    - mapped theme_id(s)
    - mapped feature_ids (if applicable)
    - rank (overall)
- Top 3 needs summary
- Top 3 pain points summary

Optional Fields

- Segment differences (per persona/tier) | OPTIONAL
- Opportunity notes (what to build/change) | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Every need/pain point must map to at least one theme_id from URD-02.
- Evidence pointers must exist for high-ranked items (top 5).
- Feature mapping is optional at this stage; if present, only use existing IDs from PRD-04.
- Do not translate directly into requirements here; recommendations live in URD-02 and changes are tracked via STK decisions.

Output Format

1) Ranked Needs & Pain Points (canonical)

item_id	type	statement	persona_ids	severity_or_importance	frequency_signal	rank	theme_ids	evidence_pointers	mapped_feature_ids	notes
n_01	need	{{items[0].statement}}	{{items[0].persona_ids}}	{{items[0].severity}}	{{items[0].frequency}}	{{items[0].rank}}	{{items[0].theme_ids}}	{{items[0].evidence}}	{{items[0].feature_ids}}	{{items[0].notes}}

p_01	pain_point	{{items[1].statement}}	{{items[1].persona_ids}}	{{items[1].severity}}	{{items[1].frequency}}	{{items[1].rank}}	{{items[1].theme_ids}}	{{items[1].evidence}}	{{items[1].feature_ids}}	{{items[1].notes}}

2) Top Needs Summary (required)

1. {{top_needs[0]}}
2. {{top_needs[1]}}
3. {{top_needs[2]}}

3) Top Pain Points Summary (required)

1. {{top_pains[0]}}
2. {{top_pains[1]}}
3. {{top_pains[2]}}

4) Persona/Segment Differences (optional)

- {{segment_differences[0]}} | OPTIONAL
- {{segment_differences[1]}} | OPTIONAL

5) Opportunity Notes (optional)

- {{opportunities[0]}} | OPTIONAL
- {{opportunities[1]}} | OPTIONAL

6) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:URD-02}}, {{xref:PRD-03}} | OPTIONAL
- Downstream: {{xref:PRD-04}} | OPTIONAL, {{xref:DES-01}} | OPTIONAL, {{xref:PRD-09}} | OPTIONAL, {{xref:RSC-03}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Minimum 5 needs + 5 pain points; keep evidence pointers simple.
- intermediate: Required. Rank items and map to themes; add persona coverage.
- advanced: Required. Add opportunity notes and stronger mapping to feature IDs where possible.

Unknown Handling

- UNKNOWN_ALLOWED: `mapped_feature_ids`, `segment_differences`, `opportunities`, `open_questions`, `frequency_signal` (qualitative)
- If any top-5 ranked item lacks evidence pointers → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESEARCH
- Pass conditions:
  - required_fields_present == true
  - needs_count >= 5
  - pain_points_count >= 5
  - every_item_has_theme_mapping == true
  - top5_have_evidence == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

URD-04

URD-04 — Journey Map (current vs target)

Header Block

- template_id: URD-04
- title: Journey Map (current vs target)
- type: user_research
- template_version: 1.0.0
- output_path: 10_app/research/URD-04_Journey_Map.md
- compliance_gate_id: TMP-05.PRIMARY.RESEARCH
- upstream_dependencies: ["URD-02", "URD-03"]
- inputs_required: ["URD-02", "URD-03", "PRD-03", "PRD-04", "GLOSSARY", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": false}

Purpose

Document the user journey in two states—current and target—to clarify steps, emotions/friction, and opportunities for intervention. This provides a bridge between research findings and design flows without specifying UI layout or implementation.

Inputs Required

- URD-02: {{xref:URD-02}}
- URD-03: {{xref:URD-03}}
- PRD-03: {{xref:PRD-03}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- GLOSSARY: {{glossary.terms}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Persona(s) covered (at least 1)
- Journey scope (which user goal / scenario)
- Journey phases (3–10)
- For each phase (current and target):
    - user goal
    - user actions
    - touchpoints/channels

    - pain points/friction
    - evidence pointer(s)
    - opportunity notes
- Summary of top frictions (top 3–7)
- Summary of target improvements (top 3–7)

Optional Fields

- Emotional curve (qualitative) | OPTIONAL
- Ownership/actor mapping (who does what) | OPTIONAL
- Mapped feature IDs | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Every major friction should map to URD-02 themes and URD-03 items when possible.
- Do not define screens/routes; that belongs in DES templates.
- If mapping to features, only use existing IDs from PRD-04.
- Evidence pointers must exist for any claimed pain point.

Output Format

1) Journey Setup

- **Persona(s):** {{journey.persona_ids}}
- **Scenario / goal:** {{journey.scenario}}
- **Scope boundaries:** {{journey.scope}} | OPTIONAL
- **Channels/touchpoints in scope:** {{journey.channels}} | OPTIONAL

2) Journey Map — Current State (required)

phase_id	phase_name	user_goal	user_actions	touchpoints	friction/pain	evidence_pointers	opportunity_notes	emotion_signal
ph_01	{{current[0].phase_name}}	{{current[0].goal}}	{{current[0].actions}}	{{current[0].touchpoints}}	{{current[0].friction}}	{{current[0].evidence}}	{{current[0].opportunity}}	{{current[0].emotion}}

ph_02	{{current[1].phase_name}}	{{current[1].goal}}	{{current[1].actions}}	{{current[1].touchpoints}}	{{current[1].friction}}	{{current[1].evidence}}	{{current[1].opportunity}}	{{current[1].emotion}}

3) Journey Map — Target State (required)

phase_id	phase_name	user_goal	user_actions	touchpoints	reduced_friction	enabling_changes	mapped_feature_ids	emotion_signal
ph_01	{{target[0].phase_name}}	{{target[0].goal}}	{{target[0].actions}}	{{target[0].touchpoints}}	{{target[0].reduced_friction}}	{{target[0].enablers}}	{{target[0].feature_ids}}	{{target[0].emotion}}
ph_02	{{target[1].phase_name}}	{{target[1].goal}}	{{target[1].actions}}	{{target[1].touchpoints}}	{{target[1].reduced_friction}}	{{target[1].enablers}}	{{target[1].feature_ids}}	{{target[1].emotion}}

4) Top Frictions (required)

1. {{top_frictions[0]}} (maps to: {{top_frictions[0].theme_ids}} / {{top_frictions[0].item_ids}}) | OPTIONAL
2. {{top_frictions[1]}} | OPTIONAL
3. {{top_frictions[2]}} | OPTIONAL

5) Target Improvements (required)

1. {{target_improvements[0]}}
2. {{target_improvements[1]}}
3. {{target_improvements[2]}}

6) Actor/Ownership Mapping (optional)

- {{ownership[0]}} | OPTIONAL

7) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:URD-02}}, {{xref:URD-03}}, {{xref:PRD-03}} | OPTIONAL
- Downstream: {{xref:DES-01}}, {{xref:PRD-04}} | OPTIONAL, {{xref:RSC-03}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 3–5 phases, 1 persona, current+target tables filled.
- intermediate: Required. Add evidence pointers and map frictions to themes/items.
- advanced: Not required. (Advanced system translation happens in DES flows and MAP templates.)

Unknown Handling

- UNKNOWN_ALLOWED: `emotion_signal`, `ownership`, `mapped_feature_ids`, `channels`, `open_questions`
- If any phase includes a pain/friction claim with no evidence_pointers → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESEARCH
- Pass conditions:
  - required_fields_present == true
  - phases_count >= 3
  - current_and_target_present == true
  - friction_claims_have_evidence == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

URD-05

URD-05 — Validation Plan (what to test, how to measure)

Header Block

- template_id: URD-05
- title: Validation Plan (what to test, how to measure)
- type: user_research
- template_version: 1.0.0
- output_path: 10_app/research/URD-05_Validation_Plan.md
- compliance_gate_id: TMP-05.PRIMARY.RESEARCH
- upstream_dependencies: ["URD-02", "URD-03", "PRD-02"]
- inputs_required: ["URD-02", "URD-03", "PRD-02", "PRD-04", "SMIP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define what must be validated (assumptions, usability, value, feasibility), how it will be tested, and what measurement signals will confirm or falsify the hypotheses. This is the bridge from discovery into build prioritization, metrics instrumentation, and release readiness.

Inputs Required

- URD-02: {{xref:URD-02}}
- URD-03: {{xref:URD-03}}
- PRD-02: {{xref:PRD-02}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- SMIP-01: {{xref:SMIP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing experiment notes: {{inputs.experiment_notes}} | OPTIONAL

Required Fields

- Validation goals (2–8)
- Hypotheses to validate (3–15)
- For each hypothesis:
    - hypothesis_id

    - statement
    - type (value/usability/feasibility/risk)
    - mapped needs/pain points (URD-03 item_ids)
    - mapped goals/metrics (PRD-02 metric_ids if available)
    - method (prototype test, A/B, pilot, survey, etc.)
    - success criteria (pass/fail)
    - measurement signals (quant/qual)
    - sample target (or **UNKNOWN**)
    - timeline
    - owner
- Decision rules (what happens if pass/fail)
- Risks/limitations

Optional Fields

- Tooling | OPTIONAL
- Variants / experiment design notes | OPTIONAL
- Guardrails | OPTIONAL
- Open questions | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Each hypothesis must map to at least one URD-03 need/pain point item.
- Success criteria must be testable (numbers or explicit qualitative thresholds).
- If measurement requires instrumentation, reference SMIP docs.
- Do not define implementation tasks; this is validation only.

Output Format

1) Validation Overview

- **Owner:** {{validation.owner}}
- **Timeline:** {{validation.timeline}}
- **Scope:** {{validation.scope}} | OPTIONAL
- **Tools:** {{validation.tools}} | OPTIONAL

2) Validation Goals

- {{validation.goals[0]}}
- {{validation.goals[1]}}

3) Hypotheses Matrix (required)

hypothesis_id	statement	type	mapped_urd_item_ids	mapped_metric_ids	method	success_criteria	signals	sample_target	owner	timeline
vh_01	{{hypotheses[0].statement}}	{{hypotheses[0].type}}	{{hypotheses[0].urd_item_ids}}	{{hypotheses[0].metric_ids}}	{{hypotheses[0].method}}	{{hypotheses[0].success_criteria}}	{{hypotheses[0].signals}}	{{hypotheses[0].sample}}	{{hypotheses[0].owner}}	{{hypotheses[0].timeline}}
vh_02	{{hypotheses[1].statement}}	{{hypotheses[1].type}}	{{hypotheses[1].urd_item_ids}}	{{hypotheses[1].metric_ids}}	{{hypotheses[1].method}}	{{hypotheses[1].success_criteria}}	{{hypotheses[1].signals}}	{{hypotheses[1].sample}}	{{hypotheses[1].owner}}	{{hypotheses[1].timeline}}

4) Decision Rules (required)

Define what changes when hypotheses pass/fail.

rule_id	condition	decision/action	affected_docs
dr_01	{{decision_rules[0].condition}}	{{decision_rules[0].action}}	{{decision_rules[0].affected_docs}}
dr_02	{{decision_rules[1].condition}}	{{decision_rules[1].action}}	{{decision_rules[1].affected_docs}}

5) Guardrails (optional)

- {{guardrails[0]}} | OPTIONAL
- {{guardrails[1]}} | OPTIONAL

6) Risks / Limitations (required)

- {{risks[0]}}
- {{risks[1]}} | OPTIONAL

7) Variants / Experiment Notes (optional)

- {{variants[0]}} | OPTIONAL

8) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:URD-02}}, {{xref:URD-03}}, {{xref:PRD-02}} | OPTIONAL
- Downstream: {{xref:SMIP-01}} | OPTIONAL, {{xref:SMIP-02}} | OPTIONAL, {{xref:RSC-03}} | OPTIONAL, {{xref:STK-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 3–5 hypotheses, simple methods, clear pass/fail statements.
- intermediate: Required. Map hypotheses to URD items and metrics; include decision rules.
- advanced: Required. Add guardrails and tighter success criteria; link outcomes to doc updates.

Unknown Handling

- UNKNOWN_ALLOWED: `sample_target`, `tools`, `variants`, `guardrails`, `open_questions`
- If success_criteria is UNKNOWN for any hypothesis → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RESEARCH
- Pass conditions:
  - required_fields_present == true
  - hypotheses_count >= 3
  - every_hypothesis_maps_to_urd_items == true
  - every_hypothesis_has_success_criteria == true
  - decision_rules_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Stakeholders & Governance (STK)

Stakeholders & Governance (STK)

- **STK-01** Stakeholder Map (roles, decision rights)
- **STK-02** Decision Log (what/why/when)
- **STK-03** RACI / Ownership Matrix
- **STK-04** Approval Gates (who signs what, when)

STK-01

STK-01 — Stakeholder Map (roles, decision rights)

Header Block

- template_id: STK-01
- title: Stakeholder Map (roles, decision rights)
- type: stakeholders_governance
- template_version: 1.0.0
- output_path: 10_app/governance/STK-01_Stakeholder_Map.md
- compliance_gate_id: TMP-05.PRIMARY.GOV
- upstream_dependencies: ["PRD-01", "PRD-02"]
- inputs_required: ["PRD-01", "PRD-02", "PRD-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define who the stakeholders are, what decisions they influence, and what authority they have. This creates deterministic ownership and prevents approval ambiguity during design, build, and release.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-02: {{xref:PRD-02}} | OPTIONAL
- PRD-03: {{xref:PRD-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Org context: {{inputs.org_context}} | OPTIONAL

Required Fields

- Stakeholder list (minimum 3 for non-trivial products)
- For each stakeholder:
    - stakeholder_id
    - name/role (title)
    - domain/area of responsibility
    - decision rights (what they can approve)
    - communication channel

    - availability / SLA for responses
- Decision areas list (product, design, architecture, security, data, release)
- Escalation path (if conflict)

Optional Fields

- External stakeholders | OPTIONAL
- Delegates / backups | OPTIONAL
- Notes | OPTIONAL

Rules

- Stakeholder IDs must be stable and unique (`stk_<slug>`).
- Decision rights must be written as explicit scopes (not vague “approves everything”).
- If any required decision area has no owner, mark **UNKNOWN** and block completeness.
- This doc defines “who decides,” STK-02 defines “what was decided.”

Output Format

1) Stakeholders (canonical)

stakeholder_id	name_or_title	area	decision_rights	comms	response_sla	backup	notes
stk_01	{{stakeholders[0].name}}	{{stakeholders[0].area}}	{{stakeholders[0].decision_rights}}	{{stakeholders[0].comms}}	{{stakeholders[0].sla}}	{{stakeholders[0].backup}}	{{stakeholders[0].notes}}
stk_02	{{stakeholders[1].name}}	{{stakeholders[1].area}}	{{stakeholders[1].decision_rights}}	{{stakeholders[1].comms}}	{{stakeholders[1].sla}}	{{stakeholders[1].backup}}	{{stakeholders[1].notes}}

2) Decision Areas Coverage (required)

decision_area	primary_owner_stakeholder_id	backup_stakeholder_id	notes
product	{{coverage.product.owner}}	{{coverage.product.backup}}	{{coverage.product.notes}}

design	{{coverage.design.owner}}	{{coverage.design.backup}}	{{coverage.design.notes}}
architecture	{{coverage.arch.owner}}	{{coverage.arch.backup}}	{{coverage.arch.notes}}
security	{{coverage.security.owner}}	{{coverage.security.backup}}	{{coverage.security.notes}}
data	{{coverage.data.owner}}	{{coverage.data.backup}}	{{coverage.data.notes}}
release	{{coverage.release.owner}}	{{coverage.release.backup}}	{{coverage.release.notes}}

3) Escalation Path (required)

- **Escalation rule:** {{escalation.rule}}
- **Escalates to:** {{escalation.to_stakeholder_id}}
- **Time threshold:** {{escalation.time_threshold}}

4) External Stakeholders (optional)

- {{external[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-02}} | OPTIONAL, {{xref:PRD-03}} | OPTIONAL
- Downstream: {{xref:STK-02}}, {{xref:STK-03}}, {{xref:STK-04}}
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. List stakeholders + decision rights + escalation.
- intermediate: Required. Add SLAs and backup owners.
- advanced: Required. Tighten decision scopes to match approval gates.

Unknown Handling

- UNKNOWN_ALLOWED: `backup`, `notes`, `external`
- If any required decision_area has UNKNOWN owner → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.GOV
- Pass conditions:
  - required_fields_present == true
  - stakeholders_count >= 3
  - decision_area_coverage_complete == true
  - escalation_path_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

STK-02

STK-02 — Decision Log (what/why/when)

Header Block

- template_id: STK-02
- title: Decision Log (what/why/when)
- type: stakeholders_governance
- template_version: 1.0.0
- output_path: 10_app/governance/STK-02_Decision_Log.md
- compliance_gate_id: TMP-05.PRIMARY.GOV
- upstream_dependencies: ["STK-01"]
- inputs_required: ["STK-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create the canonical audit of material decisions that affect scope, architecture, security, data, UX, release, and operations. This log is the source of truth for “why we chose X,” and is used to prevent re-litigating decisions.

Inputs Required

- STK-01: {{xref:STK-01}}
- Standards: {{standards.index}} | OPTIONAL
- Decision notes: {{inputs.decision_notes}} | OPTIONAL

Required Fields

- Decision entries (can be 0, but must be explicit)
- For each decision:
    - decision_id
    - date
    - decision_title
    - decision_area (product/design/architecture/security/data/release/ops)
    - decision_statement
    - context
    - options_considered
    - rationale
    - approver_stakeholder_id

    - status (proposed/approved/reversed)
    - affected_docs (IDs)
    - reversal_pointer (if reversed)

Optional Fields

- Evidence/links | OPTIONAL
- Follow-ups | OPTIONAL

Rules

- Decision IDs must be stable and unique (`dec_YYYYMMDD_<slug>` or `dec_<seq>`).
- Approver must be a stakeholder from STK-01.
- If a decision changes PRD scope or requirements, it must reference PRD-04 and RSC-04.
- Reversals must be explicit and point to the reversing decision.

Output Format

1) Decision Log (canonical)

decision_id	date	area	title	decision_statement	approver_stakeholder_id	status	affected_docs	links
dec_01	{{decisions[0].date}}	{{decisions[0].area}}	{{decisions[0].title}}	{{decisions[0].statement}}	{{decisions[0].approver_id}}	{{decisions[0].status}}	{{decisions[0].affected_docs}}	{{decisions[0].links}}

2) Decision Detail Blocks (required per entry)

{{decisions[0].decision_id}} — {{decisions[0].title}}

- **Area:** {{decisions[0].area}}
- **Date:** {{decisions[0].date}}
- **Approver:** {{decisions[0].approver_id}}
- **Status:** {{decisions[0].status}}

Context
{{decisions[0].context}}

Options considered

- {{decisions[0].options[0]}}
- {{decisions[0].options[1]}} | OPTIONAL

Decision
{{decisions[0].statement}}

Rationale
{{decisions[0].rationale}}

Affected docs
{{decisions[0].affected_docs}}

Follow-ups

- {{decisions[0].followups[0]}} | OPTIONAL

Reversal (if reversed)

- reversal_pointer: {{decisions[0].reversal_pointer}} | OPTIONAL

Cross-References

- Upstream: {{xref:STK-01}}
- Downstream: {{xref:RSC-04}} | OPTIONAL, {{xref:PRD-04}} | OPTIONAL, {{xref:ARC-*}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Capture decisions + rationale + approver.
- intermediate: Required. Include options considered and affected docs.
- advanced: Required. Maintain reversals and follow-ups with traceability.

Unknown Handling

- UNKNOWN_ALLOWED: `links`, `followups`, `options_considered` (minimum 1 option still required)
- If status == approved and approver_stakeholder_id is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.GOV

- Pass conditions:
    - required_fields_present == true
    - decision_ids_unique == true
    - approved_decisions_have_approver == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

STK-03

STK-03 — RACI / Ownership Matrix

Header Block

- template_id: STK-03
- title: RACI / Ownership Matrix
- type: stakeholders_governance
- template_version: 1.0.0
- output_path: 10_app/governance/STK-03_RACI_Ownership.md
- compliance_gate_id: TMP-05.PRIMARY.GOV
- upstream_dependencies: ["STK-01"]
- inputs_required: ["STK-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": false}

Purpose

Define operational ownership across recurring responsibilities so execution is deterministic (who does what). This is the canonical “who is responsible/accountable” table used during build and release.

Inputs Required

- STK-01: {{xref:STK-01}}
- Standards: {{standards.index}} | OPTIONAL

Required Fields

- Responsibility list (minimum 10 for non-trivial products)
- For each responsibility:
    - responsibility_id
    - area
    - description
    - R (responsible) stakeholder_id(s)
    - A (accountable) stakeholder_id
    - C (consulted) stakeholder_id(s)
    - I (informed) stakeholder_id(s)

Optional Fields

- SLA/hand-off notes | OPTIONAL
- Backup owners | OPTIONAL

Rules

- A must be exactly one stakeholder.
- Stakeholder IDs must come from STK-01.
- Responsibilities must cover at least: product, design, backend, frontend, data, security, ops, QA, release.

Output Format

1) RACI Matrix (canonical)

responsibility_id	area	description	R	A	C	I	notes
resp_01	{{rows[0].area}}	{{rows[0].description}}	{{rows[0].R}}	{{rows[0].A}}	{{rows[0].C}}	{{rows[0].I}}	{{rows[0].notes}}

2) Coverage Checklist (required)

- product covered: {{coverage.product}}
- design covered: {{coverage.design}}
- backend covered: {{coverage.backend}}
- frontend covered: {{coverage.frontend}}
- data covered: {{coverage.data}}
- security covered: {{coverage.security}}
- ops covered: {{coverage.ops}}
- qa covered: {{coverage.qa}}
- release covered: {{coverage.release}}

Cross-References

- Upstream: {{xref:STK-01}}
- Downstream: {{xref:STK-04}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Minimum responsibilities + R/A assigned.
- intermediate: Required. Add C/I lists and coverage checklist.

- advanced: Not required. (Advanced ops workflows live in OPS/REL.)

Unknown Handling

- UNKNOWN_ALLOWED: `notes`, `backup_owners`
- If any responsibility has A == UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.GOV
- Pass conditions:
  - required_fields_present == true
  - responsibilities_count >= 10
  - all_rows_have_A == true
  - coverage_checklist_complete == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

STK-04

STK-04 — Approval Gates (who signs what, when)

Header Block

- template_id: STK-04
- title: Approval Gates (who signs what, when)
- type: stakeholders_governance
- template_version: 1.0.0
- output_path: 10_app/governance/STK-04_Approval_Gates.md
- compliance_gate_id: TMP-05.PRIMARY.GOV
- upstream_dependencies: ["STK-01", "STK-03"]
- inputs_required: ["STK-01", "STK-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the approval checkpoints for the kit so builds can be gated deterministically. This specifies which docs/artifacts require sign-off, by whom, and under what pass conditions.

Inputs Required

- STK-01: {{xref:STK-01}}
- STK-03: {{xref:STK-03}} | OPTIONAL
- Standards: {{standards.index}} | OPTIONAL

Required Fields

- Gate list (minimum 5 for non-trivial products)
- For each gate:
  - gate_id
  - name
  - stage (requirements/design/architecture/implementation/security/qa/release)
  - required_artifacts (doc IDs)
  - approver_stakeholder_id(s)
  - pass_conditions (human-verifiable)
  - evidence_required (links/proof pointers)
  - SLA (approval time expectation)

    - failure_path (what happens if rejected)

Optional Fields

- Delegated approvals | OPTIONAL
- Conditional gates (only for certain targets) | OPTIONAL

Rules

- Approvers must exist in STK-01.
- Gates must align to the pipeline's gating model; do not invent gate semantics that conflict with TMP-05.
- Each gate must declare evidence_required; otherwise it is not enforceable.
- If a gate references artifacts that don't exist, mark **UNKNOWN** and block completeness.

Output Format

1) Approval Gates (canonical)

gate_id	name	stage	required_artifacts	approver_ids	pass_conditions	evidence_required	sla	failure_path
gate_01	{{gates[0].name}}	{{gates[0].stage}}	{{gates[0].artifacts}}	{{gates[0].approvers}}	{{gates[0].pass_conditions}}	{{gates[0].evidence}}	{{gates[0].sla}}	{{gates[0].failure_path}}
gate_02	{{gates[1].name}}	{{gates[1].stage}}	{{gates[1].artifacts}}	{{gates[1].approvers}}	{{gates[1].pass_conditions}}	{{gates[1].evidence}}	{{gates[1].sla}}	{{gates[1].failure_path}}

2) Gate Coverage by Stage (required)

- requirements gates: {{coverage.requirements}}
- design gates: {{coverage.design}}
- architecture gates: {{coverage.architecture}}
- implementation gates: {{coverage.implementation}}
- security gates: {{coverage.security}}
- qa gates: {{coverage.qa}}
- release gates: {{coverage.release}}

3) Escalation for Stuck Gates (required)

- **Escalate after:** {{stuck.escalate_after}}
- **Escalate to:** {{stuck.escalate_to_stakeholder_id}}
- **Method:** {{stuck.method}}

Cross-References

- Upstream: {{xref:STK-01}}, {{xref:STK-03}} | OPTIONAL
- Downstream: {{xref:TRC-03}} | OPTIONAL, {{xref:TRC-04}} | OPTIONAL, {{xref:REL-01}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define gates + approvers + required artifacts.
- intermediate: Required. Add evidence requirements and failure paths.
- advanced: Required. Add conditional gates and tighter pass conditions aligned to release criteria.

Unknown Handling

- UNKNOWN_ALLOWED: `delegated_approvals`, `conditional_gates`
- If any gate has UNKNOWN approver_ids or UNKNOWN required_artifacts → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.GOV
- Pass conditions:
  - required_fields_present == true
  - gates_count >= 5
  - all_gates_have_approvers == true
  - all_gates_have_evidence == true
  - coverage_by_stage_complete == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Domain Model & Glossary (DMG)

Domain Model & Glossary (DMG)

- **DMG-01** Domain Glossary (canonical terms)
- **DMG-02** Concept Model (entities + relationships, narrative)
- **DMG-03** Invariants & Definitions (must-always-be-true rules)
- **DMG-04** Event Vocabulary (canonical events/actions)

DMG-01

DMG-01 — Domain Glossary (canonical terms)

Header Block

- template_id: DMG-01
- title: Domain Glossary (canonical terms)
- type: domain_model_glossary
- template_version: 1.0.0
- output_path: 10_app/domain/DMG-01_Domain_Glossary.md
- compliance_gate_id: TMP-05.PRIMARY.DOMAIN
- upstream_dependencies: ["PRD-01", "PRD-03"]
- inputs_required: ["PRD-01", "PRD-03", "PRD-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Create the canonical vocabulary for the product domain so all docs, APIs, schemas, UI copy, and tests use consistent meanings. This is the authority for naming and definitions (not implementation).

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-03: {{xref:PRD-03}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing terminology notes: {{inputs.term_notes}} | OPTIONAL

Required Fields

- Term list (minimum 20 for non-trivial products; justify if smaller)
- For each term:
    - term_id
    - term
    - definition (1–3 sentences)
    - synonyms (if any)
    - anti-definition (what it is NOT) | OPTIONAL

    - related terms
    - canonical usage examples (1–3)
    - owner (who can change definition)
- Naming rules (basic, product-specific)
- Deprecated terms list (if any)

Optional Fields

- Acronyms list | OPTIONAL
- External references | OPTIONAL
- Open questions | OPTIONAL

Rules

- Definitions must not conflict with PRD and DMG-02; if conflict exists, log in PRD-08 and/or STK-02.
- Prefer one canonical term per concept; synonyms must point to canonical term_id.
- Deprecations must include replacement term_id.
- Terms used in IDs (feature/entity/endpoint) should match canonical term spellings where possible.

Output Format

1) Glossary (canonical)

term_id	term	definition	synonyms	not_this	related_term_ids	usage_examples	owner	status
t_001	{{terms[0].term}}	{{terms[0].definition}}	{{terms[0].synonyms}}	{{terms[0].not_this}}	{{terms[0].related_ids}}	{{terms[0].examples}}	{{terms[0].owner}}	{{terms[0].status}} (active/deprecated)
t_002	{{terms[1].term}}	{{terms[1].definition}}	{{terms[1].synonyms}}	{{terms[1].not_this}}	{{terms[1].related_ids}}	{{terms[1].examples}}	{{terms[1].owner}}	{{terms[1].status}}

2) Naming Rules (required)

- **Preferred casing:** {{naming.casing}}
- **Singular vs plural rules:** {{naming.singular_plural}}
- **Forbidden terms (if any):** {{naming.forbidden}} | OPTIONAL
- **UI copy rules (if any):** {{naming.ui_copy_rules}} | OPTIONAL

3) Deprecated Terms (optional)

old_term_id	old_term	replacement_term_id	reason	deprecated_on
{{deprecated[0].old_id}}	{{deprecated[0].old_term}}	{{deprecated[0].new_id}}	{{deprecated[0].reason}}	{{deprecated[0].date}}

4) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL
- {{open_questions[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-03}} | OPTIONAL, {{xref:PRD-04}} | OPTIONAL
- Downstream: {{xref:DMG-02}}, {{xref:DATA-01}} | OPTIONAL, {{xref:API-01}} | OPTIONAL, {{xref:FE-*}} | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define core nouns/verbs; keep definitions short.
- intermediate: Required. Add synonyms and anti-definitions for ambiguous terms.
- advanced: Required. Maintain deprecations and ownership.

Unknown Handling

- UNKNOWN_ALLOWED: `synonyms`, `not_this`, `related_term_ids`, `usage_examples`, `deprecated_terms`, `open_questions`
- If a term is used widely but has UNKNOWN definition → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DOMAIN
- Pass conditions:

- required_fields_present == true
- terms_count >= 20 (or justified)
- no_conflicting_definitions == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

DMG-02

DMG-02 — Concept Model (entities + relationships, narrative)

Header Block

- template_id: DMG-02
- title: Concept Model (entities + relationships, narrative)
- type: domain_model_glossary
- template_version: 1.0.0
- output_path: 10_app/domain/DMG-02_Concept_Model.md
- compliance_gate_id: TMP-05.PRIMARY.DOMAIN
- upstream_dependencies: ["DMG-01", "PRD-04"]
- inputs_required: ["DMG-01", "PRD-04", "PRD-06", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Describe the domain concept model at a logical level: core entities, their responsibilities, and how they relate—without committing to database schemas. This anchors DATA-01/02 and ARC-02.

Inputs Required

- DMG-01: {{xref:DMG-01}}
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing model notes: {{inputs.model_notes}} | OPTIONAL

Required Fields

- Entity list (logical, 5–30)
- For each entity:
    - entity_id
    - name (matches glossary term where applicable)
    - description
    - key attributes (logical)
    - key relationships (to other entities)

    - lifecycle states (optional but preferred)
- Relationship list (with cardinality)
- Narrative walkthrough of 2–5 core workflows using entities

Optional Fields

- Bounded contexts / subdomains | OPTIONAL
- Derived events | OPTIONAL
- Open questions | OPTIONAL

Rules

- Logical model only: no DB column types, no migration details.
- Entity names should align with DMG-01 canonical terms.
- Relationships must include direction + cardinality (1:1, 1:N, N:M).
- If a workflow references an entity not defined, add it or mark **UNKNOWN** and block completeness.

Output Format

1) Entities (logical)

entity_id	name	description	key_attributes	lifecycle_states	notes
ent_01	{{entities[0].name}}	{{entities[0].description}}	{{entities[0].attributes}}	{{entities[0].states}}	{{entities[0].notes}}
ent_02	{{entities[1].name}}	{{entities[1].description}}	{{entities[1].attributes}}	{{entities[1].states}}	{{entities[1].notes}}

2) Relationships (logical)

rel_id	from_entity_id	to_entity_id	cardinality	description
rel_01	{{rels[0].from}}	{{rels[0].to}}	{{rels[0].cardinality}}	{{rels[0].description}}
rel_02	{{rels[1].from}}	{{rels[1].to}}	{{rels[1].cardinality}}	{{rels[1].description}}

3) Workflow Narratives (required)

wf_01 — {{workflows[0].name}}

- Steps:
    1. {{workflows[0].steps[0]}}
    2. {{workflows[0].steps[1]}}
- Entities involved: {{workflows[0].entity_ids}}
- Outputs/artifacts: {{workflows[0].outputs}} | OPTIONAL

4) Bounded Contexts (optional)

- {{bounded_contexts[0]}} | OPTIONAL

5) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DMG-01}}, {{xref:PRD-04}} | OPTIONAL
- Downstream: {{xref:DATA-01}}, {{xref:DATA-02}}, {{xref:ARC-02}}
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define entities + relationships; keep narratives short.
- intermediate: Required. Add lifecycle states and workflow narratives.
- advanced: Required. Add bounded contexts and clearer invariants linkage.

Unknown Handling

- UNKNOWN_ALLOWED: `lifecycle_states`, `bounded_contexts`, `derived_events`, `open_questions`
- If any relationship references UNKNOWN entity → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DOMAIN
- Pass conditions:
    - required_fields_present == true
    - entities_count >= 5
    - relationships_present == true
    - workflows_count >= 2
    - placeholder_resolution == true

    - no_unapproved_unknowns == true

DMG-03 — Invariants & Definitions (must-always-be-true rules)

Header Block

- template_id: DMG-03
- title: Invariants & Definitions (must-always-be-true rules)
- type: domain_model_glossary
- template_version: 1.0.0
- output_path: 10_app/domain/DMG-03_Invariants_Definitions.md
- compliance_gate_id: TMP-05.PRIMARY.DOMAIN
- upstream_dependencies: ["DMG-02", "BRP-01"]
- inputs_required: ["DMG-02", "BRP-01", "PRD-06", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the non-negotiable domain truths (invariants) and formal definitions that must hold across all implementations. These rules anchor validation, database constraints, authorization, and test assertions.

Inputs Required

- DMG-02: {{xref:DMG-02}}
- BRP-01: {{xref:BRP-01}} | OPTIONAL
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Invariants list (minimum 10 for non-trivial products)
- For each invariant:
    - inv_id
    - statement (must/never)
    - scope (entity/relationship/system)
    - related_entity_ids
    - related_business_rule_ids (optional)

    - enforcement points (API/DB/UI/ops)
    - test strategy (how to verify)
    - severity (hard/soft)
- Definitions list (if any must be formalized beyond glossary)

Optional Fields

- Exception cases | OPTIONAL
- Open questions | OPTIONAL

Rules

- Invariants must be testable (can be asserted).
- If an invariant conflicts with BRP rules, escalate to STK-02 decision.
- “Hard” invariants must declare at least one enforcement point and a test strategy.

Output Format

1) Invariants (canonical)

inv_id	**statement**	**scope**	**entity_ids**	**br_rule_ids**	**enforcement_points**	**test_strategy**	**severity**	**exceptions**
inv_01	{{invariants[0].statement}}	{{invariants[0].scope}}	{{invariants[0].entity_ids}}	{{invariants[0].br_ids}}	{{invariants[0].enforcement}}	{{invariants[0].test_strategy}}	{{invariants[0].severity}}	{{invariants[0].exceptions}}
inv_02	{{invariants[1].statement}}	{{invariants[1].scope}}	{{invariants[1].entity_ids}}	{{invariants[1].br_ids}}	{{invariants[1].enforcement}}	{{invariants[1].test_strategy}}	{{invariants[1].severity}}	{{invariants[1].exceptions}}

2) Formal Definitions (optional)

def_id	term_id	formal_definition	notes
def_01	{{defs[0].term_id}}	{{defs[0].definition}}	{{defs[0].notes}}

3) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DMG-02}}, {{xref:BRP-01}} | OPTIONAL
- Downstream: {{xref:DATA-03}}, {{xref:API-02}}, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Capture 10 core invariants; simple enforcement points.
- intermediate: Required. Add test strategies and severity.
- advanced: Required. Tighten enforcement mapping to DB/API/UI and document exceptions.

Unknown Handling

- UNKNOWN_ALLOWED: `br_rule_ids`, `exceptions`, `formal_definitions`, `open_questions`
- If severity == hard and enforcement_points is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DOMAIN
- Pass conditions:
  - required_fields_present == true
  - invariants_count >= 10
  - hard_invariants_have_enforcement_and_tests == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DMG-04 — Event Vocabulary (canonical events/actions)

Header Block

- template_id: DMG-04
- title: Event Vocabulary (canonical events/actions)
- type: domain_model_glossary
- template_version: 1.0.0
- output_path: 10_app/domain/DMG-04_Event_Vocabulary.md
- compliance_gate_id: TMP-05.PRIMARY.DOMAIN
- upstream_dependencies: ["DMG-02", "PRD-04"]
- inputs_required: ["DMG-02", "PRD-04", "BRP-01", "SMIP-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical event/action names used across analytics, audit logs, workflows, notifications, and (if applicable) event-driven architecture. This prevents naming drift and enables deterministic mapping (SMIP, OBS, AUDIT, EVT/WEBHOOKS).

Inputs Required

- DMG-02: {{xref:DMG-02}}
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- BRP-01: {{xref:BRP-01}} | OPTIONAL
- SMIP-02: {{xref:SMIP-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Event list (minimum 15 for non-trivial products)
- For each event:
    - event_id
    - event_name (canonical)
    - trigger (what causes it)
    - actor (who/what emits)
    - related_entity_ids
    - required properties (payload fields)
    - optional properties

    - consumers (who uses it: analytics/notifications/audit/etc.)
    - retention/audit requirements (if any)
- Naming conventions for events

Optional Fields

- Versioning strategy | OPTIONAL
- Deprecations | OPTIONAL
- Open questions | OPTIONAL

Rules

- Event names must be stable and consistent (`noun.verb` or `verb_noun`—choose one and stick).
- If an event is used for analytics, it must align with SMIP-02 naming and property rules.
- If an event is used for audit/security, it must align with SEC/IAM audit requirements.
- Required properties must be concrete; if unknown, mark **UNKNOWN** and add open question.

Output Format

1) Naming Conventions (required)

- **Convention:** {{events.naming_convention}}
- **Casing:** {{events.casing}}
- **Allowed namespaces (if any):** {{events.namespaces}} | OPTIONAL

2) Event Catalog (canonical)

event_id	event_name	trigger	actor	entity_ids	required_props	optional_props	consumers	notes
ev_01	{{events.list[0].name}}	{{events.list[0].trigger}}	{{events.list[0].actor}}	{{events.list[0].entity_ids}}	{{events.list[0].required_props}}	{{events.list[0].optional_props}}	{{events.list[0].consumers}}	{{events.list[0].notes}}

ev_02	{{events.list[1].name}}	{{events.list[1].trigger}}	{{events.list[1].actor}}	{{events.list[1].entity_ids}}	{{events.list[1].required_props}}	{{events.list[1].optional_props}}	{{events.list[1].consumers}}	{{events.list[1].notes}}

3) Deprecations (optional)

old_event_id	old_event_name	replacement_event_id	reason	deprecated_on
{{deprecations[0].old_id}}	{{deprecations[0].old_name}}	{{deprecations[0].new_id}}	{{deprecations[0].reason}}	{{deprecations[0].date}}

4) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DMG-02}}, {{xref:PRD-04}} | OPTIONAL
- Downstream: {{xref:SMIP-02}} | OPTIONAL, {{xref:OBS-*}} | *OPTIONAL, {{xref:MSG-}}* | OPTIONAL, {{xref:API-02}} | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define core events and required properties at high level.
- intermediate: Required. Add consumers and align to SMIP-02.
- advanced: Required. Add deprecations/versioning strategy and tighter property definitions.

Unknown Handling

- UNKNOWN_ALLOWED: `optional_props`, `consumers`, `notes`, `versioning_strategy`, `deprecations`, `open_questions`
- If required_props is UNKNOWN for a high-impact event → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DOMAIN
- Pass conditions:
    - required_fields_present == true
    - events_count >= 15 (or justified)
    - event_names_follow_convention == true
    - placeholder_resolution == true
    - no_unapproved_unknowns == true

DMG-03

DMG-03 — Invariants & Definitions (must-always-be-true rules)

Header Block

- template_id: DMG-03
- title: Invariants & Definitions (must-always-be-true rules)
- type: domain_model_glossary
- template_version: 1.0.0
- output_path: 10_app/domain/DMG-03_Invariants_Definitions.md
- compliance_gate_id: TMP-05.PRIMARY.DOMAIN
- upstream_dependencies: ["DMG-02", "BRP-01"]
- inputs_required: ["DMG-02", "BRP-01", "PRD-06", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the non-negotiable domain truths (invariants) and formal definitions that must hold across all implementations. These rules anchor validation, database constraints, authorization, and test assertions.

Inputs Required

- DMG-02: {{xref:DMG-02}}
- BRP-01: {{xref:BRP-01}} | OPTIONAL
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Invariants list (minimum 10 for non-trivial products)
- For each invariant:
    - inv_id
    - statement (must/never)
    - scope (entity/relationship/system)
    - related_entity_ids
    - related_business_rule_ids (optional)
    - enforcement points (API/DB/UI/ops)

    - test strategy (how to verify)
    - severity (hard/soft)
- Definitions list (if any must be formalized beyond glossary)

Optional Fields

- Exception cases | OPTIONAL
- Open questions | OPTIONAL

Rules

- Invariants must be testable (can be asserted).
- If an invariant conflicts with BRP rules, escalate to STK-02 decision.
- “Hard” invariants must declare at least one enforcement point and a test strategy.

Output Format

1) Invariants (canonical)

inv_id	statement	scope	entity_ids	br_rule_ids	enforcement_points	test_strategy	severity	exceptions
inv_01	{{invariants[0].statement}}	{{invariants[0].scope}}	{{invariants[0].entity_ids}}	{{invariants[0].br_ids}}	{{invariants[0].enforcement}}	{{invariants[0].test_strategy}}	{{invariants[0].severity}}	{{invariants[0].exceptions}}
inv_02	{{invariants[1].statement}}	{{invariants[1].scope}}	{{invariants[1].entity_ids}}	{{invariants[1].br_ids}}	{{invariants[1].enforcement}}	{{invariants[1].test_strategy}}	{{invariants[1].severity}}	{{invariants[1].exceptions}}

2) Formal Definitions (optional)

def_id	term_id	formal_definition	notes
def_01	{{defs[0].term_id}}	{{defs[0].definition}}	{{defs[0].notes}}

3) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DMG-02}}, {{xref:BRP-01}} | OPTIONAL
- Downstream: {{xref:DATA-03}}, {{xref:API-02}}, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Capture 10 core invariants; simple enforcement points.
- intermediate: Required. Add test strategies and severity.
- advanced: Required. Tighten enforcement mapping to DB/API/UI and document exceptions.

Unknown Handling

- UNKNOWN_ALLOWED: `br_rule_ids`, `exceptions`, `formal_definitions`, `open_questions`
- If severity == hard and enforcement_points is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DOMAIN
- Pass conditions:
  - required_fields_present == true
  - invariants_count >= 10
  - hard_invariants_have_enforcement_and_tests == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

DMG-04

DMG-04 — Event Vocabulary (canonical events/actions)

Header Block

- template_id: DMG-04
- title: Event Vocabulary (canonical events/actions)
- type: domain_model_glossary
- template_version: 1.0.0
- output_path: 10_app/domain/DMG-04_Event_Vocabulary.md
- compliance_gate_id: TMP-05.PRIMARY.DOMAIN
- upstream_dependencies: ["DMG-02", "PRD-04"]
- inputs_required: ["DMG-02", "PRD-04", "BRP-01", "SMIP-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical event/action names used across analytics, audit logs, workflows, notifications, and (if applicable) event-driven architecture. This prevents naming drift and enables deterministic mapping (SMIP, OBS, AUDIT, EVT/WEBHOOKS).

Inputs Required

- DMG-02: {{xref:DMG-02}}
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- BRP-01: {{xref:BRP-01}} | OPTIONAL
- SMIP-02: {{xref:SMIP-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Event list (minimum 15 for non-trivial products)
- For each event:
    - event_id
    - event_name (canonical)
    - trigger (what causes it)
    - actor (who/what emits)
    - related_entity_ids

    - required properties (payload fields)
    - optional properties
    - consumers (who uses it: analytics/notifications/audit/etc.)
    - retention/audit requirements (if any)
- Naming conventions for events

Optional Fields

- Versioning strategy | OPTIONAL
- Deprecations | OPTIONAL
- Open questions | OPTIONAL

Rules

- Event names must be stable and consistent (`noun.verb` or `verb_noun`—choose one and stick).
- If an event is used for analytics, it must align with SMIP-02 naming and property rules.
- If an event is used for audit/security, it must align with SEC/IAM audit requirements.
- Required properties must be concrete; if unknown, mark **UNKNOWN** and add open question.

Output Format

1) Naming Conventions (required)

- **Convention:** {{events.naming_convention}}
- **Casing:** {{events.casing}}
- **Allowed namespaces (if any):** {{events.namespaces}} | OPTIONAL

2) Event Catalog (canonical)

event_id	event_name	trigger	actor	entity_ids	required_props	optional_props	consumers	notes
ev_	{{events.list[0].name}}	{{events.list[0].trigger}}	{{events.list[0].actor}}	{{events.list[0].entity_ids}}	{{events.list[0].required_props}}	{{events.list[0].optional_props}}	{{events.list[0].consumers}}	{{events.list[0].notes}}

ev_01	{{events.list[1].name}}	{{events.list[1].trigger}}	{{events.list[1].actor}}	{{events.list[1].entity_ids}}	{{events.list[1].required_props}}	{{events.list[1].optional_props}}	{{events.list[1].consumers}}	{{events.list[1].notes}}
ev_02								

3) Deprecations (optional)

old_event_id	old_event_name	replacement_event_id	reason	deprecated_on
{{deprecations[0].old_id}}	{{deprecations[0].old_name}}	{{deprecations[0].new_id}}	{{deprecations[0].reason}}	{{deprecations[0].date}}

4) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:DMG-02}}, {{xref:PRD-04}} | OPTIONAL
- Downstream: {{xref:SMIP-02}} | OPTIONAL, {{xref:OBS-*}}* | *OPTIONAL, {{xref:MSG-}}* | OPTIONAL, {{xref:API-02}} | OPTIONAL
- Standards: {{standards.rules[STD-NAMING]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define core events and required properties at high level.
- intermediate: Required. Add consumers and align to SMIP-02.
- advanced: Required. Add deprecations/versioning strategy and tighter property definitions.

Unknown Handling

- UNKNOWN_ALLOWED: `optional_props`, `consumers`, `notes`, `versioning_strategy`, `deprecations`, `open_questions`
- If required_props is UNKNOWN for a high-impact event → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.DOMAIN
- Pass conditions:
  - required_fields_present == true
  - events_count >= 15 (or justified)
  - event_names_follow_convention == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Roadmap & Scope Control (RSC)

Roadmap & Scope Control (RSC)
RSC-01 Release Roadmap (milestones)
RSC-02 Scope Boundaries (in/out + rationale)
RSC-03 Prioritization Framework (method + scoring)
RSC-04 Change Control Policy (how scope changes)

RSC-01

RSC-01 — Release Roadmap (milestones)

Header Block

- template_id: RSC-01
- title: Release Roadmap (milestones)
- type: roadmap_scope
- template_version: 1.0.0
- output_path: 10_app/roadmap/RSC-01_Release_Roadmap.md
- compliance_gate_id: TMP-05.PRIMARY.ROADMAP
- upstream_dependencies: ["PRD-02", "PRD-04", "PRD-06"]
- inputs_required: ["PRD-02", "PRD-04", "PRD-06", "STK-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical release roadmap: milestones, scope slices, and target dates (or date ranges). This is not a task plan; it is a product-level sequencing map that aligns stakeholders and gates.

Inputs Required

- PRD-02: {{xref:PRD-02}} | OPTIONAL
- PRD-04: {{xref:PRD-04}}
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- STK-04: {{xref:STK-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing roadmap notes: {{inputs.roadmap_notes}} | OPTIONAL

Required Fields

- Roadmap horizon (e.g., MVP, next 90 days, 2–4 releases)
- Milestones list (minimum 3)
- For each milestone:
    - milestone_id
    - name
    - objective
    - target_date (or **UNKNOWN**)
    - included feature_ids (or “TBD set” with rules)

    - exit criteria (high level)
    - dependencies (internal/external)
    - stakeholders/approvers
- Risks to roadmap (top 3–10)

Optional Fields

- Confidence level per milestone | OPTIONAL
- Beta/RC phases | OPTIONAL
- Notes | OPTIONAL

Rules

- Must align to: {{standards.rules[STD-CANONICAL-TRUTH]}} | OPTIONAL
- Milestones must reference existing feature IDs from PRD-04 where possible.
- Exit criteria must be measurable or tied to gates (QA/REL).
- Do not include implementation tasks; that belongs in IMP.

Output Format

1) Roadmap Overview

- **Horizon:** {{roadmap.horizon}}
- **Cadence:** {{roadmap.cadence}} | OPTIONAL
- **Primary stakeholders:** {{roadmap.stakeholders}} | OPTIONAL

2) Milestones (canonical)

milestone_id	name	objective	target_date	included_feature_ids	exit_criteria	dependencies	approver_ids	confidence
ms_01	{{milestones[0].name}}	{{milestones[0].objective}}	{{milestones[0].date}}	{{milestones[0].feature_ids}}	{{milestones[0].exit_criteria}}	{{milestones[0].dependencies}}	{{milestones[0].approvers}}	{{milestones[0].confidence}}
ms_02	{{milestones[1]	{{milestones[1].o	{{milestones[	{{milestones[1].fe	{{milestones[1].exit_criteria}}	{{milestones[1].dependencies}}	{{milestones[1].a	{{milestones[1].co

.name}	bjective}	1].date	ature_ids	pprovers	nfidence}
}	}	}}	}}	}}	}

3) Roadmap Risks (required)

- {{roadmap.risks[0]}}
- {{roadmap.risks[1]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-04}}, {{xref:PRD-06}} | OPTIONAL, {{xref:PRD-02}} | OPTIONAL
- Downstream: {{xref:IMP-01}} | OPTIONAL, {{xref:REL-*}} | OPTIONAL, {{xref:QA-04}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 3 milestones with objectives + feature IDs.
- intermediate: Required. Add exit criteria + dependencies.
- advanced: Required. Add confidence and approvals aligned to STK-04.

Unknown Handling

- UNKNOWN_ALLOWED: `target_date`, `confidence`, `notes`, `beta_phases`
- If any milestone has UNKNOWN objective → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.ROADMAP
- Pass conditions:
  - required_fields_present == true
  - milestones_count >= 3
  - milestones_have_objectives == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

RSC-02

RSC-02 — Scope Boundaries (in/out + rationale)

Header Block

- template_id: RSC-02
- title: Scope Boundaries (in/out + rationale)
- type: roadmap_scope
- template_version: 1.0.0
- output_path: 10_app/roadmap/RSC-02_Scope_Boundaries.md
- compliance_gate_id: TMP-05.PRIMARY.SCOPE
- upstream_dependencies: ["PRD-01", "PRD-04"]
- inputs_required: ["PRD-01", "PRD-04", "STK-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Lock the explicit boundaries of what is in-scope vs out-of-scope for the current build phase, with rationale. This prevents scope creep and serves as a gating reference for change control.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- STK-02: {{xref:STK-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Current phase label (MVP/Release 1/etc.)
- In-scope list (minimum 10 statements OR all P0 features)
- Out-of-scope list (minimum 10 statements)
- Rationale per item (brief)
- Boundary rules (how to interpret ambiguous cases)
- Escalation path (who decides) (links to STK)

Optional Fields

- Future scope (later) | OPTIONAL
- Known trade-offs | OPTIONAL
- Open questions | OPTIONAL

Rules

- “In-scope” can be features, capabilities, platforms, or constraints—must be explicit.
- If using feature IDs, reference PRD-04 only.
- Out-of-scope items must be specific enough to be testable.

Output Format

1) Phase

- **Phase:** {{scope.phase}}
- **Definition:** {{scope.phase_definition}} | OPTIONAL

2) In-Scope (required)

scope_id	statement	type (feature/capability/platform)	rationale	references
in_01	{{in_scope[0].statement}}	{{in_scope[0].type}}	{{in_scope[0].rationale}}	{{in_scope[0].refs}}
in_02	{{in_scope[1].statement}}	{{in_scope[1].type}}	{{in_scope[1].rationale}}	{{in_scope[1].refs}}

3) Out-of-Scope (required)

scope_id	statement	rationale	revisit_trigger
out_01	{{out_scope[0].statement}}	{{out_scope[0].rationale}}	{{out_scope[0].revisit_trigger}}
out_02	{{out_scope[1].statement}}	{{out_scope[1].rationale}}	{{out_scope[1].revisit_trigger}}

4) Boundary Rules (required)

- {{boundary_rules[0]}}
- {{boundary_rules[1]}} | OPTIONAL

5) Escalation / Decision Path (required)

- **Owner:** {{scope.owner_stakeholder_id}}
- **Approver:** {{scope.approver_stakeholder_id}} | OPTIONAL
- **Decision log:** {{xref:STK-02}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-04}} | OPTIONAL
- Downstream: {{xref:RSC-04}}, {{xref:IMP-01}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Clear in/out lists + owner.
- intermediate: Required. Add boundary rules + revisit triggers.
- advanced: Required. Add trade-offs and references.

Unknown Handling

- UNKNOWN_ALLOWED: `future_scope`, `tradeoffs`, `open_questions`, `references`
- If owner_stakeholder_id is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SCOPE
- Pass conditions:
  - required_fields_present == true
  - in_scope_present == true
  - out_scope_present == true
  - boundary_rules_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

RSC-03

RSC-03 — Prioritization Framework (method + scoring)

Header Block

- template_id: RSC-03
- title: Prioritization Framework (method + scoring)
- type: roadmap_scope
- template_version: 1.0.0
- output_path: 10_app/roadmap/RSC-03_Prioritization_Framework.md
- compliance_gate_id: TMP-05.PRIMARY.SCOPE
- upstream_dependencies: ["PRD-04", "URD-03", "PRD-02"]
- inputs_required: ["PRD-04", "URD-03", "PRD-02", "RISK-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": false}

Purpose

Define the repeatable method used to prioritize features and scope decisions (e.g., RICE/ICE/WSJF/custom). This makes prioritization explainable, consistent, and auditable.

Inputs Required

- PRD-04: {{xref:PRD-04}}
- URD-03: {{xref:URD-03}} | OPTIONAL
- PRD-02: {{xref:PRD-02}} | OPTIONAL
- RISK-02: {{xref:RISK-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Selected method name
- Score factors (3–8)
- Scoring rules (scale definitions)
- Weighting (if any)
- Tie-break rules
- Example scored items (minimum 5 features)

Optional Fields

- Persona/segment weighting | OPTIONAL
- Risk adjustment | OPTIONAL
- Notes | OPTIONAL

Rules

- If scoring features, only use feature IDs from PRD-04.
- The framework must be usable without subjective ambiguity (define scales).
- If weights are used, they must sum to 1.0 or 100.

Output Format

1) Method

- **Method name:** {{prior.method}}
- **Why this method:** {{prior.rationale}} | OPTIONAL

2) Factors & Scales (required)

factor_id	name	description	scale	weight
f_01	{{factors[0].name}}	{{factors[0].desc}}	{{factors[0].scale}}	{{factors[0].weight}}
f_02	{{factors[1].name}}	{{factors[1].desc}}	{{factors[1].scale}}	{{factors[1].weight}}

3) Tie-break Rules (required)

- {{prior.tie_break[0]}}
- {{prior.tie_break[1]}} | OPTIONAL

4) Scored Features (required, min 5)

feature_id	feature_name	factor_scores	total_score	notes
{{spec.features_by_id[feat_x].id}}	{{spec.features_by_id[feat_x].name}}	{{scores[feat_x].factors}}	{{scores[feat_x].total}}	{{scores[feat_x].notes}}

Cross-References

- Upstream: {{xref:PRD-04}}, {{xref:URD-03}} | OPTIONAL
- Downstream: {{xref:RSC-01}}, {{xref:IMP-01}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Simple method with defined scales.
- intermediate: Required. Add weights and tie-break rules.
- advanced: Not required. (Advanced optimization belongs in planning/analytics.)

Unknown Handling

- UNKNOWN_ALLOWED: `weights`, `persona_weighting`, `risk_adjustment`, `notes`
- If scales are undefined → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SCOPE
- Pass conditions:
  - required_fields_present == true
  - factors_defined == true
  - scales_defined == true
  - scored_items_count >= 5
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

RSC-04

RSC-04 — Change Control Policy (how scope changes)

Header Block

- template_id: RSC-04
- title: Change Control Policy (how scope changes)
- type: roadmap_scope
- template_version: 1.0.0
- output_path: 10_app/roadmap/RSC-04_Change_Control_Policy.md
- compliance_gate_id: TMP-05.PRIMARY.SCOPE
- upstream_dependencies: ["RSC-02", "STK-04"]
- inputs_required: ["RSC-02", "STK-04", "STK-02", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the policy for proposing, evaluating, approving, and recording scope changes. This prevents silent scope creep and ensures changes are traceable to decisions and roadmap updates.

Inputs Required

- RSC-02: {{xref:RSC-02}}
- STK-04: {{xref:STK-04}} | OPTIONAL
- STK-02: {{xref:STK-02}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Change request types (bugfix, scope add, scope removal, requirement change, timeline change)
- Required information for a change request
- Evaluation criteria (impact, risk, cost, schedule, security, UX)
- Approval rules (who signs off)
- Decision recording rules (must log to STK-02)
- Update rules (which docs must be updated when change is approved)
- Emergency change path (hotfix / security)

Optional Fields

- SLA for change review | OPTIONAL
- Limits (max changes per milestone) | OPTIONAL

Rules

- Approved changes must create a STK-02 decision entry.
- Any scope change must update PRD-04 and/or RSC-01 as applicable.
- Emergency path must still record after-the-fact decision within defined SLA.

Output Format

1) Change Request Schema (required)

field	required	description
change_id	true	{{change.schema.change_id}}
requester	true	{{change.schema.requester}}
type	true	{{change.schema.type}}
description	true	{{change.schema.description}}
rationale	true	{{change.schema.rationale}}
impacted_docs	true	{{change.schema.impacted_docs}}
impacted_feature_ids	false	{{change.schema.feature_ids}}
risk_level	true	{{change.schema.risk_level}}
proposed_timeline_change	false	{{change.schema.timeline_change}}

2) Evaluation Criteria (required)

- {{change.eval[0]}}
- {{change.eval[1]}}

3) Approval Rules (required)

- **Approvers by type:** {{change.approvers_by_type}}
- **Default approver:** {{change.default_approver}} | OPTIONAL

- **Approval gates reference:** {{xref:STK-04}} | OPTIONAL

4) Recording & Update Rules (required)

- **Record decision in:** {{xref:STK-02}}
- **Update required docs:** {{change.update_docs}}
- **Notify stakeholders:** {{change.notify}} | OPTIONAL

5) Emergency Change Path (required)

- **When allowed:** {{change.emergency.when_allowed}}
- **Who approves:** {{change.emergency.approver}}
- **Retroactive logging SLA:** {{change.emergency.logging_sla}}

Cross-References

- Upstream: {{xref:RSC-02}}, {{xref:STK-04}} | OPTIONAL
- Downstream: {{xref:STK-02}}, {{xref:RSC-01}}, {{xref:PRD-04}}
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Simple schema + approval + logging rules.
- intermediate: Required. Add evaluation criteria and update rules.
- advanced: Required. Add emergency path with strict retroactive logging.

Unknown Handling

- UNKNOWN_ALLOWED: `review_sla`, `limits`, `notify`
- If approval rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.SCOPE
- Pass conditions:
  - required_fields_present == true
  - change_request_schema_present == true
  - approval_rules_present == true
  - recording_rules_present == true
  - emergency_path_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Risk & Assumptions (RISK)

Risk & Assumptions (RISK)
RISK-01 Assumptions Register
RISK-02 Risk Register (probability/impact/mitigation)
RISK-03 Dependency Map (external/internal)
RISK-04 Contingency Triggers (what causes plan changes)

RISK-01

RISK-01 — Assumptions Register

Header Block

- template_id: RISK-01
- title: Assumptions Register
- type: risk_assumptions
- template_version: 1.0.0
- output_path: 10_app/risk/RISK-01_Assumptions_Register.md
- compliance_gate_id: TMP-05.PRIMARY.RISK
- upstream_dependencies: ["PRD-01", "PRD-02", "PRD-06", "URD-01"]
- inputs_required: ["PRD-01", "PRD-02", "PRD-06", "URD-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Maintain the canonical list of assumptions the product/build depends on, including validation plans and timelines. Assumptions are not constraints; they are beliefs that must be tested or monitored.

Inputs Required

- PRD-01: {{xref:PRD-01}}
- PRD-02: {{xref:PRD-02}} | OPTIONAL
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- URD-01: {{xref:URD-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Assumptions list (minimum 10 for non-trivial products)
- For each assumption:
    - assumption_id
    - statement
    - category (market/user/tech/ops/legal/financial)
    - why_it_matters
    - risk_if_false
    - validation_plan
    - validate_by (date or milestone) (or **UNKNOWN**)

- owner
- status (unvalidated/validated/invalidated/monitoring)
- evidence (if validated) | OPTIONAL
- impacted_feature_ids / domains / docs

Optional Fields

- Monitoring signals | OPTIONAL
- Open questions | OPTIONAL

Rules

- Each assumption must have a validation_plan (no exceptions).
- If status is validated/invalidated, evidence must be present.
- If validate_by is UNKNOWN, include a mitigation note.

Output Format

1) Assumptions Register (canonical)

assumption_id	statement	category	why_it_matters	risk_if_false	validation_plan	validate_by	owner	status	evidence	impacted_refs
a_01	{{assumptions[0].statement}}	{{assumptions[0].category}}	{{assumptions[0].why}}	{{assumptions[0].risk}}	{{assumptions[0].plan}}	{{assumptions[0].validate_by}}	{{assumptions[0].owner}}	{{assumptions[0].status}}	{{assumptions[0].evidence}}	{{assumptions[0].impacted_refs}}
a_02	{{assumptions[1].statement}}	{{assumptions[1].category}}	{{assumptions[1].why}}	{{assumptions[1].risk}}	{{assumptions[1].plan}}	{{assumptions[1].validate_by}}	{{assumptions[1].owner}}	{{assumptions[1].status}}	{{assumptions[1].evidence}}	{{assumptions[1].impacted_refs}}

2) Status Summary (required)

- unvalidated: {{summary.unvalidated_count}}

- validated: {{summary.validated_count}}
- invalidated: {{summary.invalidated_count}}
- monitoring: {{summary.monitoring_count}}

3) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-01}}, {{xref:PRD-02}} | OPTIONAL, {{xref:PRD-06}} | OPTIONAL, {{xref:URD-01}} | OPTIONAL
- Downstream: {{xref:URD-05}} | OPTIONAL, {{xref:RISK-02}}, {{xref:STK-02}} | OPTIONAL, {{xref:IMP-01}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 10 assumptions with validation plans.
- intermediate: Required. Add validate_by + impacted_refs.
- advanced: Required. Add monitoring signals and evidence trails.

Unknown Handling

- UNKNOWN_ALLOWED: `validate_by` (must include mitigation), `monitoring_signals`, `open_questions`
- If validation_plan is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RISK
- Pass conditions:
  - required_fields_present == true
  - assumptions_count >= 10
  - every_assumption_has_validation_plan == true
  - validated_or_invalidated_have_evidence == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

RISK-02

RISK-02 — Risk Register (probability/impact/mitigation)

Header Block

- template_id: RISK-02
- title: Risk Register (probability/impact/mitigation)
- type: risk_assumptions
- template_version: 1.0.0
- output_path: 10_app/risk/RISK-02_Risk_Register.md
- compliance_gate_id: TMP-05.PRIMARY.RISK
- upstream_dependencies: ["RISK-01", "PRD-06", "RSC-01"]
- inputs_required: ["RISK-01", "PRD-06", "RSC-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Track and prioritize risks with clear mitigation, owners, and triggers. This is the canonical list used for planning tradeoffs, security posture, ops readiness, and release gating.

Inputs Required

- RISK-01: {{xref:RISK-01}} | OPTIONAL
- PRD-06: {{xref:PRD-06}} | OPTIONAL
- RSC-01: {{xref:RSC-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Risks list (minimum 10 for non-trivial products)
- For each risk:
    - risk_id
    - statement
    - category (product/tech/security/data/ops/legal)
    - probability (low/med/high or 1–5)
    - impact (low/med/high or 1–5)
    - severity (derived or explicit)
    - mitigation_plan

- owner
- trigger_signals (what indicates it's happening)
- status (open/mitigating/accepted/closed)
- impacted_feature_ids / domains / docs

Optional Fields

- Contingency plan | OPTIONAL
- Residual risk | OPTIONAL
- Links/evidence | OPTIONAL

Rules

- Every risk must have a mitigation_plan.
- Any "accepted" risk must include rationale and approver (stakeholder_id) or STK-02 reference.
- Top risks (by severity) must have triggers.

Output Format

1) Risk Register (canonical)

risk_id	statement	category	probability	impact	severity	mitigation_plan	trigger_signals	owner	status	impacted_refs	notes
r_01	{{risks[0].statement}}	{{risks[0].category}}	{{risks[0].prob}}	{{risks[0].impact}}	{{risks[0].severity}}	{{risks[0].mitigation}}	{{risks[0].triggers}}	{{risks[0].owner}}	{{risks[0].status}}	{{risks[0].impacted_refs}}	{{risks[0].notes}}

2) Top Risks (required)

- {{derive:TOP_RISKS(risks, 5)}} | OPTIONAL

3) Accepted Risks (optional)

risk_id	rationale	approver_or_decision_ref

{{accepted[0].risk_id}} {{accepted[0].rationale}} {{accepted[0].approval_ref}}

Cross-References

- Upstream: {{xref:RISK-01}} | OPTIONAL, {{xref:PRD-06}} | OPTIONAL, {{xref:RSC-01}} | OPTIONAL
- Downstream: {{xref:IMP-01}} | OPTIONAL, {{xref:RELIA-01}} | OPTIONAL, {{xref:SEC-01}} | OPTIONAL, {{xref:STK-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 10 risks with mitigation + owner.
- intermediate: Required. Add triggers and impacted refs.
- advanced: Required. Add accepted risk approvals and residual risk notes.

Unknown Handling

- UNKNOWN_ALLOWED: `contingency_plan`, `residual_risk`, `links`, `notes`
- If mitigation_plan is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RISK
- Pass conditions:
  - required_fields_present == true
  - risks_count >= 10
  - every_risk_has_mitigation == true
  - top_risks_have_triggers == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

RISK-03

RISK-03 — Dependency Map (external/internal)

Header Block

- template_id: RISK-03
- title: Dependency Map (external/internal)
- type: risk_assumptions
- template_version: 1.0.0
- output_path: 10_app/risk/RISK-03_Dependency_Map.md
- compliance_gate_id: TMP-05.PRIMARY.RISK
- upstream_dependencies: ["RSC-01", "IMP-01"]
- inputs_required: ["RSC-01", "PRD-04", "IMP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": false}

Purpose

List the internal and external dependencies that affect delivery, along with owners, timelines, and failure modes. This makes sequencing and risk mitigation deterministic.

Inputs Required

- RSC-01: {{xref:RSC-01}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- IMP-01: {{xref:IMP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Dependency list (minimum 8 for non-trivial products)
- For each dependency:
    - dep_id
    - name
    - type (internal_team/external_vendor/system/library/legal/infra)
    - description
    - needed_for (milestone_id / feature_id / doc_id)
    - owner
    - due_date (or **UNKNOWN**)

    - status (not_started/in_progress/blocked/done)
    - failure_mode (what happens if late/broken)
    - mitigation (fallback plan)

Optional Fields

- Contract/SLA info | OPTIONAL
- Links | OPTIONAL
- Open questions | OPTIONAL

Rules

- Every dependency must have a failure_mode and mitigation.
- If due_date is UNKNOWN, it must include a tracking plan.

Output Format

1) Dependency Map (canonical)

dep_id	name	type	needed_for	owner	due_date	status	failure_mode	mitigation	links
dep_01	{{deps[0].name}}	{{deps[0].type}}	{{deps[0].needed_for}}	{{deps[0].owner}}	{{deps[0].due_date}}	{{deps[0].status}}	{{deps[0].failure_mode}}	{{deps[0].mitigation}}	{{deps[0].links}}

2) Critical Dependencies (required)

- {{derive:CRITICAL_DEPS(deps)}} | OPTIONAL

Cross-References

- Upstream: {{xref:RSC-01}} | OPTIONAL, {{xref:IMP-01}} | OPTIONAL, {{xref:PRD-04}} | OPTIONAL
- Downstream: {{xref:RISK-02}} | OPTIONAL, {{xref:REL-01}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. List deps + needed_for + mitigation.
- intermediate: Required. Add failure modes and status tracking.
- advanced: Not required. (Advanced vendor ops lives in COMP/OPS.)

Unknown Handling

- UNKNOWN_ALLOWED: `due_date`, `links`, `contract_sla`, `open_questions`
- If failure_mode or mitigation is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RISK
- Pass conditions:
  - required_fields_present == true
  - deps_count >= 8
  - every_dep_has_failure_and_mitigation == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

RISK-04

RISK-04 — Contingency Triggers (what causes plan changes)

Header Block

- template_id: RISK-04
- title: Contingency Triggers (what causes plan changes)
- type: risk_assumptions
- template_version: 1.0.0
- output_path: 10_app/risk/RISK-04_Contingency_Triggers.md
- compliance_gate_id: TMP-05.PRIMARY.RISK
- upstream_dependencies: ["RISK-02", "RSC-01", "RELIA-01"]
- inputs_required: ["RISK-02", "RSC-01", "RELIA-01", "STK-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define explicit trigger conditions that force a plan change (scope, timeline, architecture, rollout). This makes escalation and contingency behavior deterministic instead of reactive.

Inputs Required

- RISK-02: {{xref:RISK-02}}
- RSC-01: {{xref:RSC-01}} | OPTIONAL
- RELIA-01: {{xref:RELIA-01}} | OPTIONAL
- STK-04: {{xref:STK-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Trigger list (minimum 8)
- For each trigger:
    - trig_id
    - related_risk_id
    - condition (clear threshold)
    - detection_method (how we know)
    - response_action (what changes)
    - decision_owner (stakeholder_id)

    - response_sla
    - rollback_or_mitigation_path
    - comms_required (who must be informed)

Optional Fields

- Severity tiers | OPTIONAL
- Playbook link | OPTIONAL
- Notes | OPTIONAL

Rules

- Conditions must be measurable (thresholds, counts, time windows).
- Decision owner must exist in STK-01/04 governance.
- Any trigger that changes scope must reference RSC-04 change control path.

Output Format

1) Contingency Triggers (canonical)

trig_id	related_risk_id	condition	detection_method	response_action	decision_owner	response_sla	mitigation_or_rollback	comms_required	links
trig_01	{{triggers[0].risk_id}}	{{triggers[0].condition}}	{{triggers[0].detect}}	{{triggers[0].action}}	{{triggers[0].owner}}	{{triggers[0].sla}}	{{triggers[0].rollback}}	{{triggers[0].comms}}	{{triggers[0].links}}

2) Escalation Rules (required)

- **Escalate when:** {{escalation.when}}
- **Escalate to:** {{escalation.to}}
- **Path:** {{xref:STK-04}} | OPTIONAL

Cross-References

- Upstream: {{xref:RISK-02}}, {{xref:RSC-01}} | OPTIONAL, {{xref:RELIA-01}} | OPTIONAL

- Downstream: {{xref:RSC-04}} | OPTIONAL, {{xref:REL-04}} | OPTIONAL, {{xref:OPS-05}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. 8 triggers with clear conditions + owners.
- intermediate: Required. Add response SLAs and comms rules.
- advanced: Required. Tie triggers to rollback paths and change control.

Unknown Handling

- UNKNOWN_ALLOWED: `links`, `notes`, `severity_tiers`
- If condition or response_action is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.RISK
- Pass conditions:
  - required_fields_present == true
  - triggers_count >= 8
  - every_trigger_has_condition_and_action == true
  - owners_present == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Business Rules & Policy (BRP)

Business Rules & Policy (BRP)
BRP-01 Business Rules Catalog (by ID)
BRP-02 Eligibility & Entitlement Rules
BRP-03 Pricing/Permission Policy Rules (if applicable)
BRP-04 Exceptions & Edge-Case Policy

BRP-01

BRP-01 — Business Rules Catalog (by ID)

Header Block

- template_id: BRP-01
- title: Business Rules Catalog (by ID)
- type: business_rules_policy
- template_version: 1.0.0
- output_path: 10_app/policy/BRP-01_Business_Rules_Catalog.md
- compliance_gate_id: TMP-05.PRIMARY.POLICY
- upstream_dependencies: ["PRD-04", "DMG-01", "DMG-03"]
- inputs_required: ["PRD-04", "DMG-01", "DMG-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical, testable business rules that govern system behavior. These rules are referenced by API authorization, data constraints, UI validation, and test cases. This is the “rule source,” not the implementation.

Inputs Required

- PRD-04: {{xref:PRD-04}} | OPTIONAL
- DMG-01: {{xref:DMG-01}} | OPTIONAL
- DMG-03: {{xref:DMG-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Existing policy notes: {{inputs.policy_notes}} | OPTIONAL

Required Fields

- Rule list (minimum 15 for non-trivial products)
- For each rule:
    - br_id
    - name
    - rule_statement (must/never/only if)
    - category (eligibility/entitlement/pricing/limits/workflow/data)
    - scope (system/feature/entity/endpoint)
    - related_feature_ids
    - related_entity_ids

- inputs (what the rule evaluates)
- outputs/effects (what changes)
- exceptions (if any)
- enforcement_points (UI/API/DB/ops)
- testability_notes (how to verify)
- priority (P0/P1/P2)
- status (active/deprecated)

Optional Fields

- Source references (policy docs) | OPTIONAL
- Versioning notes | OPTIONAL
- Open questions | OPTIONAL

Rules

- Rules must be testable; vague language must be converted into measurable conditions.
- Rule IDs must be stable and unique (`br_<slug>`).
- If a rule is “hard” (P0), it must declare at least one enforcement_point and testability_notes.
- If a rule conflicts with another rule or invariant, escalate to STK-02.

Output Format

1) Business Rules Catalog (canonical)

br_id	name	category	rule_statement	scope	feature_ids	entity_ids	inputs	outputs/effects	exceptions	enforcement_points	testability_notes	priority	status	notes
br_01	{{rules[0].name}}	{{rules[0].category}}	{{rules[0].statement}}	{{rules[0].scope}}	{{rules[0].feature_ids}}	{{rules[0].entity_ids}}	{{rules[0].inputs}}	{{rules[0].outputs}}	{{rules[0].exceptions}}	{{rules[0].enforcement}}	{{rules[0].testability}}	{{rules[0].priority}}	{{rules[0].status}}	{{rules[0].notes}}
br_	{{rules[1].	{{rules[1].cat	{{rules[1].stat	{{rules[1].s	{{rules[1].featur	{{rules[1].entit	{{rules[1].i	{{rules[1].ou	{{rules[1].exc	{{rules[1].enforc	{{rules[1].test	{{rules[1].pri	{{rules[1].s	{{rules[1].

02	name}}	egory}}	ement}}	cope}}	e_ids}}	y_ids}}	nputs}}	tputs}}	eptions}}	ement}}	ability}}	ority}}	tatus}}	notes}}

2) P0 Rules Summary (required)

- {{derive:LIST_P0_RULES(rules)}} | OPTIONAL

3) Conflicts (required if any)

- {{conflicts[0]}} | OPTIONAL

4) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-04}} | OPTIONAL, {{xref:DMG-03}} | OPTIONAL
- Downstream: {{xref:API-02}} | OPTIONAL, {{xref:DATA-03}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL, {{xref:IAM-03}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Capture rule_statement + category + enforcement points at high level.
- intermediate: Required. Add inputs/outputs and testability notes.
- advanced: Required. Tighten exceptions and map to features/entities.

Unknown Handling

- UNKNOWN_ALLOWED: `exceptions`, `source_references`, `notes`, `open_questions`
- If priority == P0 and enforcement_points is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.POLICY
- Pass conditions:
  - required_fields_present == true

- rules_count >= 15
- p0_rules_have_enforcement_and_testability == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

BRP-02

BRP-02 — Eligibility & Entitlement Rules

Header Block

- template_id: BRP-02
- title: Eligibility & Entitlement Rules
- type: business_rules_policy
- template_version: 1.0.0
- output_path: 10_app/policy/BRP-02_Eligibility_Entitlements.md
- compliance_gate_id: TMP-05.PRIMARY.POLICY
- upstream_dependencies: ["BRP-01", "PRD-03"]
- inputs_required: ["BRP-01", "PRD-03", "PRD-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Specify who is eligible for what and what entitlements they receive (features, limits, permissions). This is the canonical source for gating capabilities (often used by IAM, PAY/REVOPS, and UI/UX).

Inputs Required

- BRP-01: {{xref:BRP-01}}
- PRD-03: {{xref:PRD-03}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Plan/tier notes: {{inputs.tier_notes}} | OPTIONAL

Required Fields

- Eligibility criteria list (minimum 5)
- Entitlement catalog (minimum 10 entitlements)
- Mapping matrix:
    - role/persona/tier → entitlements
- For each eligibility rule:
    - elig_id
    - statement
    - evaluated_inputs
    - decision_output (eligible/ineligible + reason)

    - enforcement_points
    - exceptions
- For each entitlement:
    - ent_id
    - entitlement_name
    - description
    - scope (feature/limit/permission)
    - related_feature_ids
    - limit_value (if applicable)
    - enforcement_points
    - audit_event (if applicable)

Optional Fields

- Trial rules | OPTIONAL
- Grace periods | OPTIONAL
- Open questions | OPTIONAL

Rules

- Any eligibility/entitlement must be enforceable at API level (even if also enforced in UI).
- If tiers exist, entitlements must be deterministic by tier.
- Limit entitlements must declare units (e.g., per day, per month, per org).

Output Format

1) Eligibility Rules (canonical)

elig_id	statement	evaluated_inputs	output	enforcement_points	exceptions	notes
elig_01	{{elig[0].statement}}	{{elig[0].inputs}}	{{elig[0].output}}	{{elig[0].enforcement}}	{{elig[0].exceptions}}	{{elig[0].notes}}

2) Entitlements Catalog (canonical)

ent_id	name	scope	description	feature_ids	limit_value	units	enforcement_points	audit_event	notes

ent_01	{{ents[0].name}}	{{ents[0].scope}}	{{ents[0].desc}}	{{ents[0].feature_ids}}	{{ents[0].limit}}	{{ents[0].units}}	{{ents[0].enforcement}}	{{ents[0].audit_event}}	{{ents[0].notes}}

3) Eligibility/Entitlement Mapping (required)

subject_type (role/tier/persona)	subject_id	granted_entitlement_ids	notes
role	{{map[0].subject_id}}	{{map[0].ent_ids}}	{{map[0].notes}}

4) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:BRP-01}}, {{xref:PRD-03}} | OPTIONAL
- Downstream: {{xref:IAM-03}} | OPTIONAL, {{xref:REVOPS-01}} | OPTIONAL, {{xref:API-04}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define entitlements + who gets them.
- intermediate: Required. Add enforcement points and units for limits.
- advanced: Required. Add audit_event mapping and exception handling.

Unknown Handling

- UNKNOWN_ALLOWED: `exceptions`, `trial_rules`, `grace_periods`, `audit_event`, `open_questions`
- If any entitlement has UNKNOWN scope or enforcement_points → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.POLICY
- Pass conditions:

- required_fields_present == true
- eligibility_rules_count >= 5
- entitlements_count >= 10
- mapping_present == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

BRP-03

BRP-03 — Pricing/Permission Policy Rules (if applicable)

Header Block

- template_id: BRP-03
- title: Pricing/Permission Policy Rules
- type: business_rules_policy
- template_version: 1.0.0
- output_path: 10_app/policy/BRP-03_Pricing_Permission_Policy.md
- compliance_gate_id: TMP-05.PRIMARY.POLICY
- upstream_dependencies: ["BRP-02"]
- inputs_required: ["BRP-02", "PRD-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Capture policy rules that govern pricing, billing permissions, and paywall behavior (if the product has monetization). This informs PAY/REVOPS, UI gating, and entitlements enforcement.

Inputs Required

- BRP-02: {{xref:BRP-02}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Pricing notes: {{inputs.pricing_notes}} | OPTIONAL

Required Fields

- Policy applicability (true/false) (if false, explicitly mark N/A)
- Pricing constructs (plans, tiers, add-ons)
- Permission rules for billing actions (who can purchase/cancel/refund)
- Paywall rules (what happens when not entitled)
- Upgrade/downgrade rules (effective timing)
- Proration/refund policy pointer (if applicable)

Optional Fields

- Taxes/invoicing policy pointer | OPTIONAL
- Promotions/coupons | OPTIONAL
- Open questions | OPTIONAL

Rules

- If applicable == false, include only: rationale + references, and mark remaining sections 00_NA.
- Paywall behavior must map to entitlements from BRP-02.
- Billing permissions must map to roles from PRD-03/IAM.

Output Format

1) Applicability

- **applies:** {{pricing.applies}} (true/false)
- **rationale:** {{pricing.rationale}} | OPTIONAL

2) Pricing Constructs (if applies)

plan_id	name	included_entitlement_ids	limits	notes
plan_01	{{plans[0].name}}	{{plans[0].ent_ids}}	{{plans[0].limits}}	{{plans[0].notes}}

3) Billing Permissions (if applies)

action	allowed_role_ids	notes
purchase	{{billing_perms.purchase.roles}}	{{billing_perms.purchase.notes}}
cancel	{{billing_perms.cancel.roles}}	{{billing_perms.cancel.notes}}
refund	{{billing_perms.refund.roles}}	{{billing_perms.refund.notes}}

4) Paywall Rules (if applies)

rule_id	condition	user_experience	enforcement_points	notes
pw_01	{{paywall[0].condition}}	{{paywall[0].ux}}	{{paywall[0].enforcement}}	{{paywall[0].notes}}

5) Upgrade/Downgrade Rules (if applies)

- {{changes.upgrade_rule}}
- {{changes.downgrade_rule}} | OPTIONAL

6) N/A Marker (if not applies)

- 00_NA: {{pricing.na_block}} | OPTIONAL

7) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:BRP-02}} | OPTIONAL
- Downstream: {{xref:REVOPS-*}} | *OPTIONAL, {{xref:PAY-*}}* | OPTIONAL, {{xref:FE-*}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required unless monetization exists.
- intermediate: Required if monetization exists.
- advanced: Required if monetization exists and multiple plans/roles.

Unknown Handling

- UNKNOWN_ALLOWED: `promotions`, `tax_policy`, `open_questions`, `notes`
- If applies == true and paywall rules are UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.POLICY
- Pass conditions:
  - required_fields_present == true
  - if_applies_then_pricing_sections_complete == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

BRP-04

BRP-04 — Exceptions & Edge-Case Policy

Header Block

- template_id: BRP-04
- title: Exceptions & Edge-Case Policy
- type: business_rules_policy
- template_version: 1.0.0
- output_path: 10_app/policy/BRP-04_Exceptions_EdgeCases.md
- compliance_gate_id: TMP-05.PRIMARY.POLICY
- upstream_dependencies: ["BRP-01", "PRD-09", "DMG-03"]
- inputs_required: ["BRP-01", "PRD-09", "DMG-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical handling for edge cases and exceptions so behavior is consistent across UI/API/ops and is testable. This prevents ad-hoc “special cases” being implemented differently across the system.

Inputs Required

- BRP-01: {{xref:BRP-01}} | OPTIONAL
- PRD-09: {{xref:PRD-09}} | OPTIONAL
- DMG-03: {{xref:DMG-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL
- Known edge case notes: {{inputs.edge_notes}} | OPTIONAL

Required Fields

- Exceptions list (minimum 10 for non-trivial products)
- For each exception:
    - ex_id
    - triggering_condition
    - affected_rule_ids / invariant_ids / feature_ids
    - expected_system_behavior
    - user_experience (what user sees)
    - enforcement_points (UI/API/DB/ops)
    - logging/audit requirements

    - test cases required (tc pointers or descriptions)
    - severity (P0/P1/P2)
- Global policy for “unknown/unhandled cases”

Optional Fields

- Support playbook pointer | OPTIONAL
- Open questions | OPTIONAL

Rules

- Every exception must be testable (must map to a test case requirement).
- P0 exceptions must define logging/audit requirements and a user-facing experience.
- If “unhandled case” behavior is not defined, default must be explicit (fail-safe vs permissive).

Output Format

1) Exceptions Catalog (canonical)

ex_id	condition	affected_refs	expected_behavior	user_experience	enforcement_points	logging_audit	test_requirements	severity	notes
ex_01	{{exceptions[0].condition}}	{{exceptions[0].affected_refs}}	{{exceptions[0].behavior}}	{{exceptions[0].ux}}	{{exceptions[0].enforcement}}	{{exceptions[0].logging}}	{{exceptions[0].tests}}	{{exceptions[0].severity}}	{{exceptions[0].notes}}

2) Global Edge-Case Policy (required)

- **Default behavior:** {{policy.default_behavior}} (fail-safe / permissive / block + escalate)
- **Reason codes:** {{policy.reason_codes}} | OPTIONAL
- **Support escalation:** {{policy.support_escalation}} | OPTIONAL

3) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:BRP-01}} | OPTIONAL, {{xref:DMG-03}} | OPTIONAL
- Downstream: {{xref:API-03}} | OPTIONAL, {{xref:ARC-06}} | OPTIONAL, {{xref:QA-02}} | OPTIONAL, {{xref:ADMIN-02}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Capture top exceptions + expected behavior.
- intermediate: Required. Add enforcement points and logging requirements.
- advanced: Required. Add reason codes and explicit test requirement pointers.

Unknown Handling

- UNKNOWN_ALLOWED: `reason_codes`, `support_playbook_pointer`, `notes`, `open_questions`
- If severity == P0 and expected_behavior is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.POLICY
- Pass conditions:
  - required_fields_present == true
  - exceptions_count >= 10
  - every_exception_has_behavior_and_tests == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Success Metrics & Instrumentation Plan (SMIP)

Success Metrics & Instrumentation Plan (SMIP)
SMIP-01 KPI/OKR Definitions (targets + owners)
SMIP-02 Analytics Event Spec (event names + properties)
SMIP-03 Funnel/Conversion Definitions
SMIP-04 Experiment Measurement Plan (guardrails + success)

SMIP-01

SMIP-01 — KPI/OKR Definitions (targets + owners)

Header Block

- template_id: SMIP-01
- title: KPI/OKR Definitions (targets + owners)
- type: success_metrics_instrumentation
- template_version: 1.0.0
- output_path: 10_app/metrics/SMIP-01_KPI_OKR_Definitions.md
- compliance_gate_id: TMP-05.PRIMARY.METRICS
- upstream_dependencies: ["PRD-02"]
- inputs_required: ["PRD-02", "PRD-04", "URD-05", "STK-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical set of KPIs/OKRs with targets, owners, and measurement cadence. This is the authoritative metrics layer that downstream analytics, dashboards, experimentation, and release criteria reference.

Inputs Required

- PRD-02: {{xref:PRD-02}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- URD-05: {{xref:URD-05}} | OPTIONAL
- STK-01: {{xref:STK-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Metrics list (minimum 5)
- For each metric:
    - metric_id
    - metric_name
    - type (KPI/OKR/guardrail)
    - definition (plain language)
    - formula/logic

- target (number or threshold) (or **UNKNOWN**)
- time_horizon (MVP/30/60/90/release)
- cadence (daily/weekly/monthly)
- owner (stakeholder_id or team)
- data_source (or **UNKNOWN**)
- segments (if any)
- linked_goal_ids (from PRD-02)
- linked_feature_ids (optional)

Optional Fields

- Baseline values | OPTIONAL
- Notes | OPTIONAL
- Open questions | OPTIONAL

Rules

- Every PRD-02 goal must map to at least one metric (enforced in PRD-02).
- Metrics must be measurable; if target is UNKNOWN, include the plan for setting it (in notes).
- If data_source is UNKNOWN, it must create a SMIP-02 instrumentation question.
- Guardrails must specify stop conditions or thresholds.

Output Format

1) Metrics Catalog (canonical)

metric_id	name	type	definition	formula/logic	target	baseline	horizon	cadence	owner	data_source	segments	linked_goal_ids	linked_feature_ids	notes
met	{{metrics[0].n	{{metrics[0].t	{{metrics[0].definition}}	{{metrics[0].for	{{metrics[0].t	{{metrics[0].bas	{{metrics[0].hori	{{metrics[0].cad	{{metrics[0].o	{{metrics[0].data_so	{{metrics[0].segments}}	{{metrics[0].goal	{{metrics[0].feature	{{metrics[0].n

0	am	yp	mul	arg	elin	zon}	enc	wn	urce}	_ids	_ids}	ote
1	e}}	e}}	a}}	et}}	e}}	}	e}}	er}}	}	}}	}	s}}

2) Ownership Summary (required)

- {{derive:METRICS_BY_OWNER(metrics)}} | OPTIONAL

3) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:PRD-02}} | OPTIONAL, {{xref:URD-05}} | OPTIONAL
- Downstream: {{xref:SMIP-02}}, {{xref:SMIP-03}} | OPTIONAL, {{xref:SMIP-04}} | OPTIONAL, {{xref:OBS-05}} | OPTIONAL, {{xref:EXPER-*}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define 5 metrics with owners and cadence; targets may be UNKNOWN.
- intermediate: Required. Add formulas and data sources; reduce UNKNOWN.
- advanced: Required. Add segments, guardrails, and explicit stop thresholds.

Unknown Handling

- UNKNOWN_ALLOWED: `target`, `baseline`, `data_source`, `segments`, `notes`, `open_questions`
- If formula/logic is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.METRICS
- Pass conditions:
  - required_fields_present == true
  - metrics_count >= 5
  - every_metric_has_definition_and_formula == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

SMIP-02

SMIP-02 — Analytics Event Spec (event names + properties)

Header Block

- template_id: SMIP-02
- title: Analytics Event Spec (event names + properties)
- type: success_metrics_instrumentation
- template_version: 1.0.0
- output_path: 10_app/metrics/SMIP-02_Analytics_Event_Spec.md
- compliance_gate_id: TMP-05.PRIMARY.METRICS
- upstream_dependencies: ["SMIP-01", "DMG-04"]
- inputs_required: ["SMIP-01", "DMG-04", "PRD-04", "DGP-01", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": true}

Purpose

Define the canonical analytics event taxonomy and event properties used to measure KPIs, funnels, and experiments. This must align with privacy/data classification rules and avoid PII leakage.

Inputs Required

- SMIP-01: {{xref:SMIP-01}}
- DMG-04: {{xref:DMG-04}} | OPTIONAL
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- DGP-01: {{xref:DGP-01}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Event naming convention
- Event list (minimum 10)
- For each event:
    - event_name
    - description
    - trigger
    - actor

    - required properties (name/type)
    - optional properties (name/type)
    - PII classification per property (none/low/high) (or policy label)
    - sampling (if any)
    - destinations (warehouse, product analytics, logs)
    - linked metrics (metric_ids)
- Global required context properties (app_version, platform, user_id hash, etc.)

Optional Fields

- Event versioning/deprecations | OPTIONAL
- QA validation rules | OPTIONAL
- Open questions | OPTIONAL

Rules

- Property PII classification must align to DGP-01.
- Any user identifier must be hashed/pseudonymous unless explicitly allowed.
- Every metric that requires instrumentation must reference at least one event.
- Names must match DMG-04 convention where applicable.

Output Format

1) Global Context Properties (required)

prop_name	type	pii_class	notes
app_version	string	none	{{context.app_version_notes}}
platform	string	none	{{context.platform_notes}}

2) Event Catalog (canonical)

event_name	description	trigger	actor	required_props	optional_props	pii_class_notes	sampling	destinations	linked_metric_ids	notes
{{events[0].name}}	{{events[0].desc}}	{{events[0].trigger}}	{{events[0].actor}}	{{events[0].required_props}}	{{events[0].optional_props}}	{{events[0].pii_notes}}	{{events[0].sampling}}	{{events[0].destinations}}	{{events[0].metric_ids}}	{{events[0].notes}}

3) Deprecations (optional)

old_event_name	replacement_event_name	deprecated_on	reason
{{deprecations[0].old}}	{{deprecations[0].new}}	{{deprecations[0].date}}	{{deprecations[0].reason}}

4) QA Validation Rules (optional)

- {{qa_rules[0]}} | OPTIONAL

5) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:SMIP-01}}, {{xref:DMG-04}} | OPTIONAL
- Downstream: {{xref:SMIP-03}}, {{xref:SMIP-04}} | OPTIONAL, {{xref:BI-*}} | OPTIONAL, {{xref:OBS-01}} | OPTIONAL
- Standards: {{standards.rules[STD-PRIVACY]}} | OPTIONAL, {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define 10 events + core props; keep PII notes explicit.
- intermediate: Required. Link events to metrics and destinations.
- advanced: Required. Add sampling, deprecations, and QA validation rules.

Unknown Handling

- UNKNOWN_ALLOWED: `sampling`, `destinations`, `deprecations`, `qa_rules`, `open_questions`
- If any required_prop has UNKNOWN pii_class → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.METRICS
- Pass conditions:
  - required_fields_present == true
  - events_count >= 10

- every_metric_has_event_mapping == true
- pii_classification_complete == true
- placeholder_resolution == true
- no_unapproved_unknowns == true

SMIP-03

SMIP-03 — Funnel/Conversion Definitions

Header Block

- template_id: SMIP-03
- title: Funnel/Conversion Definitions
- type: success_metrics_instrumentation
- template_version: 1.0.0
- output_path: 10_app/metrics/SMIP-03_Funnel_Definitions.md
- compliance_gate_id: TMP-05.PRIMARY.METRICS
- upstream_dependencies: ["SMIP-01", "SMIP-02"]
- inputs_required: ["SMIP-01", "SMIP-02", "PRD-04", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": true, "intermediate": true, "advanced": false}

Purpose

Define the canonical funnels and conversion events used to measure product success. Funnels must be linked to analytics events and have clear step definitions to prevent metric drift.

Inputs Required

- SMIP-01: {{xref:SMIP-01}}
- SMIP-02: {{xref:SMIP-02}}
- PRD-04: {{xref:PRD-04}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Funnel list (minimum 2)
- For each funnel:
    - funnel_id
    - name
    - purpose
    - audience/persona (optional)
    - steps (ordered), each step references event_name(s)
    - conversion definition (what counts)
    - window (time window)
    - segmentation rules (optional)
    - linked metric_ids

    - guardrails (optional)

Optional Fields

- Drop-off analysis notes | OPTIONAL
- Open questions | OPTIONAL

Rules

- Every funnel step must reference events defined in SMIP-02.
- Conversion must be measurable and unambiguous.
- Time window must be explicit.

Output Format

1) Funnel Catalog (canonical)

funnel_id	**name**	**purpose**	**steps**	**conversion_definition**	**window**	**segments**	**linked_metric_ids**	**guardrails**	**notes**
fun_01	{{funnels[0].name}}	{{funnels[0].purpose}}	{{funnels[0].steps}}	{{funnels[0].conversion}}	{{funnels[0].window}}	{{funnels[0].segments}}	{{funnels[0].metric_ids}}	{{funnels[0].guardrails}}	{{funnels[0].notes}}

2) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:SMIP-01}}, {{xref:SMIP-02}}
- Downstream: {{xref:BI-*}}* | *OPTIONAL, {{xref:EXPER-}}* | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Required. Define 2 funnels with steps + window.

- intermediate: Required. Add linked metrics and segments.
- advanced: Not required. (Advanced experimentation lives in EXPER.)

Unknown Handling

- UNKNOWN_ALLOWED: `segments`, `guardrails`, `notes`, `open_questions`
- If any funnel step references unknown event → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.METRICS
- Pass conditions:
  - required_fields_present == true
  - funnels_count >= 2
  - all_steps_reference_known_events == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

SMIP-04

SMIP-04 — Experiment Measurement Plan (guardrails + success)

Header Block

- template_id: SMIP-04
- title: Experiment Measurement Plan (guardrails + success)
- type: success_metrics_instrumentation
- template_version: 1.0.0
- output_path: 10_app/metrics/SMIP-04_Experiment_Measurement_Plan.md
- compliance_gate_id: TMP-05.PRIMARY.METRICS
- upstream_dependencies: ["SMIP-01", "URD-05"]
- inputs_required: ["SMIP-01", "URD-05", "SMIP-03", "STANDARDS_INDEX"]
- required_by_skill_level: {"beginner": false, "intermediate": true, "advanced": true}

Purpose

Define how experiments will be measured: success metrics, guardrails, stop conditions, and analysis requirements. This is the product-facing measurement plan that can be executed by the experimentation system.

Inputs Required

- SMIP-01: {{xref:SMIP-01}}
- URD-05: {{xref:URD-05}} | OPTIONAL
- SMIP-03: {{xref:SMIP-03}} | OPTIONAL
- STANDARDS_INDEX: {{standards.index}} | OPTIONAL

Required Fields

- Applicability (true/false) (if no experiments planned, mark N/A)
- Experiment list (minimum 1 if applicable)
- For each experiment:
    - exp_id
    - hypothesis
    - primary_metric_id
    - secondary_metric_ids
    - guardrail_metric_ids

    - success threshold
    - stop conditions
    - segments
    - duration / sample size plan (or **UNKNOWN**)
    - analysis notes

Optional Fields

- Statistical method notes | OPTIONAL
- Open questions | OPTIONAL

Rules

- If applies == false, include N/A block only.
- Primary/guardrail metrics must exist in SMIP-01.
- Stop conditions must be explicit and measurable.

Output Format

1) Applicability

- **applies:** {{exper.applies}} (true/false)
- **rationale:** {{exper.rationale}} | OPTIONAL

2) Experiment Plans (if applies)

exp_id	hypothesis	primary_metric_id	guardrail_metric_ids	success_threshold	stop_conditions	segments	duration	analysis_notes
exp_01	{{exper.list[0].hypothesis}}	{{exper.list[0].primary_metric}}	{{exper.list[0].guardrails}}	{{exper.list[0].threshold}}	{{exper.list[0].stop}}	{{exper.list[0].segments}}	{{exper.list[0].duration}}	{{exper.list[0].analysis}}

3) N/A Marker (if not applies)

- 00_NA: {{exper.na_block}} | OPTIONAL

4) Open Questions (optional)

- {{open_questions[0]}} | OPTIONAL

Cross-References

- Upstream: {{xref:SMIP-01}}, {{xref:URD-05}} | OPTIONAL
- Downstream: {{xref:EXPER-*}} | OPTIONAL
- Standards: {{standards.rules[STD-UNKNOWN-HANDLING]}} | OPTIONAL

Skill Level Requiredness Rules

- beginner: Not required unless experiments planned.
- intermediate: Required if applies.
- advanced: Required if applies.

Unknown Handling

- UNKNOWN_ALLOWED: `duration`, `segments`, `analysis_notes`, `open_questions`
- If applies == true and success_threshold is UNKNOWN → block Completeness Gate.

Completeness Gate

- Gate ID: TMP-05.PRIMARY.METRICS
- Pass conditions:
  - required_fields_present == true
  - if_applies_then_experiments_present == true
  - metrics_refs_valid == true
  - placeholder_resolution == true
  - no_unapproved_unknowns == true

Domains

AXION domain list (v1)

A) Core build domains

1. `architecture`
2. `frontend`
3. `backend`
4. `mobile`
5. `integrations`
6. `api_gateway`

B) Data domains

7. `data`
8. `data_platform`
9. `indexing`
10. `search`
11. `caching`

C) Identity & admin domains

12. `identity_access`
13. `authn_authz`
14. `permissions_policy`

15. `admin_internal_tools`

D) Realtime & communications domains

16. `realtime`
17. `messaging`
18. `eventing_webhooks`
19. `notifications`

E) Monetization domains

20. `payments`
21. `billing`
22. `subscriptions_entitlements`
23. `pricing_plans`

F) Media domains

24. `media_storage`
25. `file_processing`
26. `uploads_downloads`
27. `cdn_delivery`

G) Experience domains

28. `ux`

29. `design_system`
30. `content_design`
31. `accessibility`

H) Platform operations domains

32. `ops`
33. `infrastructure`
34. `cicd_pipelines`
35. `environment_management`
36. `release`

I) Quality & reliability domains

37. `quality`
38. `test_infrastructure`
39. `test_data_environments`
40. `performance`
41. `reliability_resilience`
42. `incident_response`

J) Security, privacy, compliance domains

43. `security`

44. `telemetry_observability`
45. `logging_tracing`
46. `monitoring_alerting`
47. `slo_sla_error_budgets`
48. `compliance`
49. `privacy_data_governance`
50. `audit_forensics`
51. `risk_management`
52. `migration_import_export`
53. `backup_restore_dr`